CHAPTER 8

Harmonic Vibrations

8.1 Introduction

In giving a microscopic description of the nuclear properties, until now we have used only static independent particle models. We have seen that by a proper definition of the particles or quasi-particles and by using an effective interaction, we are able within such a picture to explain the basic properties of the groundstates of many nuclei. Spectra of nuclear excitations are also very important for an understanding of the nuclear structure. If we perform an analysis of such spectra within the nuclear shell model or within the more elaborated independent particle models such as Hartree–Fock or Hartree–Fock–Bogoliubov, we find that a series of excited states can be very adequately explained by such models as *ph*- or two-quasi-particle excitations. But there are also many excited states with features that cannot be reproduced within the framework of shell model excitations, in spite of the fact that by introducing sophisticated methods involving the breaking of symmetries we are able to take into account important correlations among the nucleons.

If, for example, we restrict ourselves to spherically symmetric doubly magic nuclei like ^{16}O, ^{40}Ca, or ^{208}Pb, then the lowest excitations according to the harmonic oscillator approximation of the nuclear shell model should lie at about $\hbar\omega_0$. For ^{16}O the experimental shell difference between $1p$ and $2s$–$1d$ shells is roughly 11.5 MeV. In the spectrum of ^{16}O, however, we see amongst other states a ($J^\pi = 3^-$, $T = 0$)-state at 6.14 MeV, and around 22

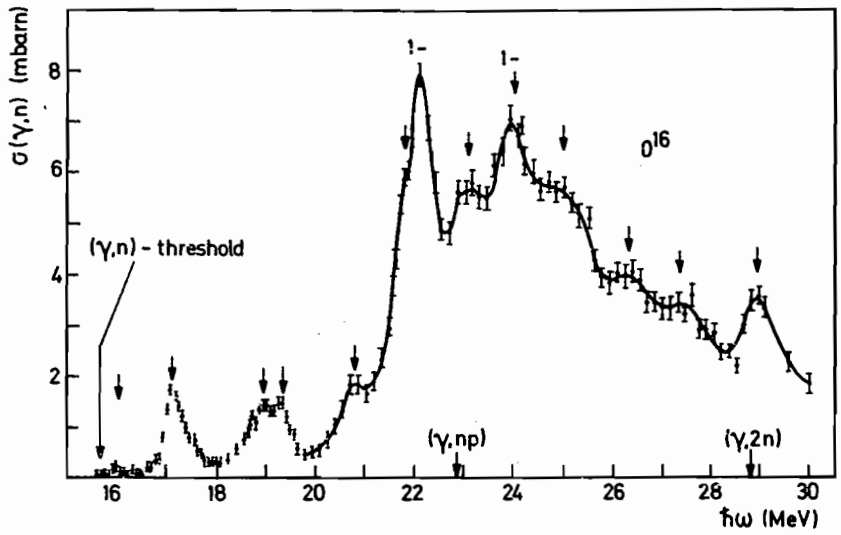


Figure 8.1. The cross section for (γ, n) processes as a function of the γ-energy (from [BCH 64]).

MeV a 5 MeV broad resonance which is usually called the giant dipole resonance (see Fig. 8.1) having the quantum numbers $J^{\pi}=1^{-}$, $T=1$.

It can be shown (see Sec. 8.3.3) that mainly $1\hbar\omega$ excitations are involved in this resonance. Therefore, the pure shell model also fails to explain the high energy of the giant dipole resonance.

It will turn out that these excitations can only be explained if we suppose that coherent participation by many nucleons takes place in the nucleus. What we understand by "*coherent*" can at the moment only be expressed in terms of the classical liquid drop approach to the nucleus (Chap. 1): Many nucleons are believed to take part in similar fashion to the shape vibrations of the nuclear drop. We will learn below what coherent means in the quantum mechanical sense. Such collective excitations furthermore generally fulfill the following criteria.

(i) Their electromagnetic transition probabilities have a collective strength such that they are one to two orders of magnitude larger than the single-particle transitions (see Sec. 2.7.2).

(ii) They show up in the spectra of different nuclei over entire regions of the periodic table with great regularity. The giant dipole resonance, for instance, has been observed throughout the periodic table and its excitation energy varies slowly with mass number [$\propto 80A^{-1/3 \div -1/6}$, see Eq. (1.44)].

The spectra of spherical as well as deformed nuclei contain a variety of states that fulfill these criteria. We will restrict ourselves to the general methods which, within the context of the *residual interaction* Eq. (2.35)

$$V_R = \frac{1}{4} \sum_{kk',ll'} \bar{v}_{kk'll'} a_k^+ a_{k'}^+ a_{l'} a_l - V_{\mathrm{HF}}, \tag{8.1}$$

allow an explanation of the high-lying (which are usually resonances) as well as low-lying (usually bound states) collective vibrations. As indicated by Eq. (8.1), we will in the following mostly use the Hartree–Fock single-particle picture because it makes things somewhat less complicated. It should be emphasized, however, that all methods which we will present in the following sections generalize straightforwardly when using HFB quasi-particles (see Chap. 7) instead of particles. In certain cases we will also show this explicitly.

In Section 8.2, we study the interaction between *ph*-pairs based on an uncorrelated HF ground state (Tamm–Dancoff approximation TDA). In Section 8.3, different types of collective excitations are presented. Ground state correlations are taken into account in Section 8.4 within the random phase approximation (RPA) and it is shown that some problems inherent to the TDA method can be solved in this way. In Section 8.5, we show how the same equations can be derived with the formalism of linear response theory and what kind of features come into play when treating density dependent interactions. In Section 8.6 we present some numerical applications of the theory. Sum rules are an interesting means of extracting general properties of such resonances and of clarifying the relations between the quantum mechanical eigenstates of the RPA-Hamiltonian and the classical picture of surface vibrations. They are discussed in Section 8.7. Finally, Sections 8.8 and 8.9 are devoted to special extensions of the RPA, such as particle–particle RPA and quasi-particle RPA.

8.2 Tamm–Dancoff Method

8.2.1 Tamm–Dancoff Secular Equation

If we fill up the shell model potential with A nucleons up to a certain Fermi level, as shown in Fig. 8.2, then all zero, single, two, three, four,..., N particle shell model excitations form a complete orthogonal set which can be used to expand the true many-nucleon wave functions of the ground state $|0\rangle$ or the excited states $|\nu\rangle$. The exact solution of the Schrödinger equation would then be obtained by a diagonalization of H in the full shell model space or, equivalently, by a variation of the following expansion coefficients. [In the following we restrict the indices m, n (i, j) to above (below) the Fermi level.]

$$|0\rangle = C_0^0|\mathrm{HF}\rangle + \sum_{mi} C_{mi}^0 a_m^+ a_i|\mathrm{HF}\rangle + \frac{1}{4}\sum_{mnij} C_{mn,ij}^0 a_m^+ a_n^+ a_i a_j|\mathrm{HF}\rangle + \cdots \quad (8.2)$$

$$|\nu\rangle = C_0^\nu|\mathrm{HF}\rangle + \sum_{mi} C_{mi}^\nu a_m^+ a_i|\mathrm{HF}\rangle + \frac{1}{4}\sum_{mnij} C_{mn,ij}^\nu a_m^+ a_n^+ a_i a_j|\mathrm{HF}\rangle + \cdots. \quad (8.3)$$

Ideally, the determinant $|\mathrm{HF}\rangle$ would be given by filling up the N lowest single-particle levels of the Hartree–Fock potential. However, the latter is

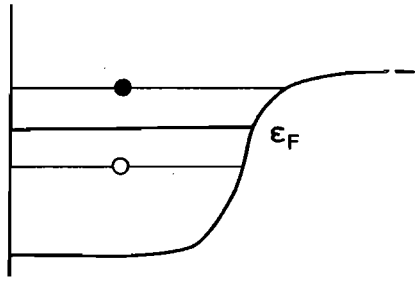


Figure 8.2. A *ph*-excitation in the shell model.

usually approximated by a phenomenological single-particle potential. The operator $a_m^+ a_i$ consequently annihilates a particle below ϵ_F (creates a hole) and creates a particle above ϵ_F. It is therefore called a particle–hole creation operator (Fig. 8.2).

The exact diagonalization of H within the full shell model space is a task which cannot be solved. Let us, however, suppose that it is reasonable in the expansion of the excited states $|\nu\rangle$ in Eq. (8.3) to go up to 1 particle–1 hole excitations only. We will see later that for certain kind of states this is indeed a very good approximation. A priori there are also reasons why this should be so. The subset of $1\hbar\omega$ particle–hole excitations are the shell model configurations lowest in energy and should therefore be important for low-lying states of negative parity. Transition probabilities of states which are excited by an electromagnetic field $F=\sum_{pp'} F_{pp'} a_p^+ a_{p'}$ are proportional to $|\langle 0|F|\nu\rangle|^2$ and these states are therefore believed to have dominant particle–hole contributions. We can thus hope to get a fair approximation for at least a certain class of states if we retain from the expansion (8.3) only the following terms

$$|\nu\rangle \simeq C_0^\nu |\mathrm{HF}\rangle + \sum_{mi} C_{mi}^\nu a_m^+ a_i |\mathrm{HF}\rangle. \tag{8.4}$$

Since $\langle \mathrm{HF}|H a_m^+ a_i|\mathrm{HF}\rangle = 0$ [Eq. (5.35)], we see that the ansatz (8.4) implies that the ground state remains a determinant, whereas it is sufficient for the excited state to retain only

$$|\nu\rangle = \sum_{mi} C_{mi}^\nu a_m^+ a_i |\mathrm{HF}\rangle. \tag{8.5}$$

The main drawback of this procedure is the fact that correlations are only taken into account for the excited states, whereas the ground state will be unchanged. Later we will see how we can build correlations into the ground state also.

Since the Hamilton operator commutes with the operator of total angular momentum **I** and in many cases also with the operator of total isospin **T**, we should use the ansatz (8.5) in a Clebsch–Gordon coupled form [see Eq. (2.46)]. In order not to spoil the simplicity of our formulae we shall omit them in the following unless they are absolutely necessary.

Using (8.5) as a variational ansatz, we obtain with

$$|\delta\nu\rangle = \sum_{mi} a_m^+ a_i |\mathrm{HF}\rangle \delta C_{mi}^{\nu} \tag{8.6}$$

as usual a secular equation for the determination of eigenvalues and expansion coefficients:

$$\sum_{nj} \{\langle \mathrm{HF}|a_i^+ a_m H a_n^+ a_j|\mathrm{HF}\rangle - E_\nu \langle \mathrm{HF}|a_i^+ a_m a_n^+ a_j|\mathrm{HF}\rangle\} C_{nj}^{\nu} = 0. \tag{8.7}$$

With (5.35), we can get a more convenient form of Eq. (8.7):

$$\sum_{nj} \langle \mathrm{HF}|a_i^+ a_m \big[H, a_n^+ a_j\big]|\mathrm{HF}\rangle C_{nj}^{\nu} = \big(E_\nu - E_0^{\mathrm{HF}}\big) C_{mi}^{\nu}. \tag{8.8}$$

The commutator in (8.8) can easily be calculated with the use of Eqs. (5.25)

$$\begin{aligned}\big[H, a_n^+ a_j\big] = &\sum_r \{t_{rn} a_r^+ a_j - t_{jr} a_n^+ a_r\} \\ &+ \frac{1}{2} \sum_{rst} \bar{v}_{rsnt} a_r^+ a_s^+ a_t a_j - \frac{1}{2} \sum_{rst} \bar{v}_{jrst} a_n^+ a_r^+ a_t a_s.\end{aligned} \tag{8.9}$$

With the definition (5.37) of the Hartree–Fock single-particle energies ϵ_p and the rules for calculating the ground state expectation values of field operators (Appendix C) we finally obtain the so-called *Tamm–Dancoff equation*:

$$\sum_{nj} \{(\epsilon_m - \epsilon_i)\delta_{mn}\delta_{ij} + \bar{v}_{mjin}\} C_{nj}^{\nu} = E_\nu^{\mathrm{TDA}} C_{mi}^{\nu}. \tag{8.10}$$

[E_0^{HF} has been set to zero by a suitable choice of the energy scale and E_ν^{TDA} is the excitation energy of $|\nu\rangle$ in Tamm–Dancoff approximation (TDA)].

For many purposes it has turned out to be useful to represent the matrix elements of the interaction $\bar{v}$ graphically (Fig. 8.3):

$$\begin{aligned}\bar{v}_{mjin} = &\int \varphi_m^*(\xi_1)\varphi_j^*(\xi_2) v(\xi_1, \xi_2) \varphi_i(\xi_1)\varphi_n(\xi_2)\, d\xi_1\, d\xi_2 \\ &- \int \varphi_m^*(\xi_1)\varphi_j^*(\xi_2) v(\xi_1, \xi_2) \varphi_n(\xi_1)\varphi_i(\xi_2)\, d\xi_1\, d\xi_2.\end{aligned} \tag{8.11}$$

The variable ξ stands symbolically for all coordinates and the integration shall include summation over spin and isospin. The rules for graphs are explained in many text books [AGD 63, Ma 67b] and since we do not want to present the graphical method as a tool by itself but rather as a pictorial representation of what we have derived analytically, we do not go into much detail here and simply give the prescription used to draw them. A wavy line shall stand for v and lines with arrows for the single-particle functions (arrow up–particle; arrow down–hole) which, according to their coordinates, are linked to the interaction points 1 and 2 of the wavy line. Furthermore, we usually agree that lines which are to the right of the interaction v in formula (8.11) point towards it, and that those lines to the left point away from it. With these conventions, $\bar{v}_{mj,\,in}$ can be displayed graphically, as shown in Fig. 8.3.

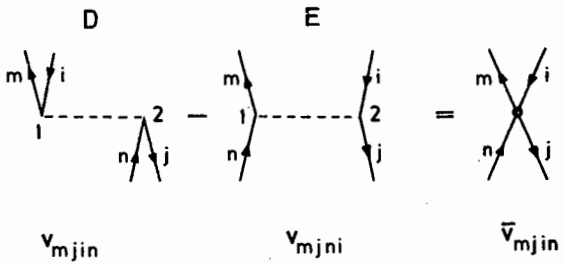


Figure 8.3. Graphical representation of the matrix elements v_{mjin} and v_{mjni}.

If we imagine a time scale perpendicular to the wavy line, a particle–hole pair nj is annihilated and another one created in the direct term D, whereas in the exchange term E a pair nj is scattered in a new one mi. Thinking of the wavy line as a pion, the meaning of "direct" and "exchange" becomes even clearer; process D being a pair creation and annihilation process (as in electrodynamics, in which the wavy line stands for the photon, the hole for the positron, and the particle for the electron). The graph E then represents a process in which a pion is exchanged between a particle–hole pair. Very often the direct plus exchange term is combined into one graph (Feynman graph), indicated in Fig. 8.3.

8.2.2 The Schematic Model

The solution of the Tamm–Dancoff equation is, in general, not obtained in the full particle–hole space, since this is a very big numerical task. Therefore, in most cases, only a "model" space is retained which includes several levels above and below ϵ_F. These calculations are also often called configuration mixing or shell model calculations. Before we go into more details, we want to study the qualitative features of the solutions of the TDA equation in a very simple model for which we can obtain the solution in an analytic form.

8.2.2.1. Solution of the TDA Equation with a Separable Interaction. The solution of Eq. (8.10) is greatly simplified if we assume that the matrix elements of the residual interaction are separable in the particle–hole indices (see Sec. 4.4.7 and [BB 59, BET 61])

$$\bar{v}_{mjin} = \lambda \cdot D_{mi} D^*_{nj}. \tag{8.12}$$

Further, it is assumed that the D_{mi} are matrix elements of a multipole operator as, for example, the quadrupole operator

$$D_{mi} = \langle m | r^2 Y_{2\mu}(\vartheta, \varphi) | i \rangle. \tag{8.13}$$

The multipolarity agrees, of course, with the angular momentum to which the particle–hole pair (m, i) is coupled. The matrix element (8.12) is certainly not antisymmetric as it should be. However, it turns out that the exchange term is small and neglecting it is a good approximation [BK 68,

Ma 74]. In general, the residual interaction is attractive for $T=0$ states (they are pushed down in energy as, for example, the 3^- $T=0$ state in ^{16}O), and repulsive for $T=1$ states [BB 59]. Therefore, we have to choose

$$\begin{aligned} &\lambda<0 \quad \text{for } T=0; \\ &\lambda>0 \quad \text{for } T=1. \end{aligned} \tag{8.14}$$

There are reasons to believe that the ansatz (8.12) for the interaction is not as bad as it might look, but here we do not want to go into the philosophy of a separable force (see Sec. 4.4.7). Rather, we wish to consider (8.12) as a convenient simplification which allows us to study the qualitative features of the TDA solution.

With the ansatz (8.12), the secular equation (8.10) has the following form.

$$(E_\nu^{\mathrm{TDA}} - \epsilon_m + \epsilon_i) C_{mi}^\nu = \lambda D_{mi} \sum_{nj} D_{nj}^* C_{nj}^\nu. \tag{8.15}$$

The states $|\nu\rangle$ should, of course, be normalized and with Eq. (8.5) we therefore have:

$$\sum_{mi} C_{mi}^{\nu*} C_{mi}^{\nu'} = \delta_{\nu\nu'}. \tag{8.16}$$

With $\sum_{nj} D_{nj}^* C_{nj}^\nu = \text{const.}$, we can solve Eq. (8.15) for the coefficients C_{mi}^ν:

$$\begin{aligned} C_{mi}^\nu &= N \cdot \frac{D_{mi}}{E_\nu^{\mathrm{TDA}} - \epsilon_m + \epsilon_i}; \\ N^{-2} &= \sum_{mi} \frac{|D_{mi}|^2}{\left(E_\nu^{\mathrm{TDA}} - \epsilon_m + \epsilon_i\right)^2} \end{aligned} \tag{8.17}$$

Multiplying Eq. (8.15) by $D_{mi}^* (E_\nu^{\mathrm{TDA}} - \epsilon_m + \epsilon_i)^{-1}$ and summing over m, i, we obtain an eigenvalue equation for the excitation energies E_ν^{TDA}:

$$\frac{1}{\lambda} = \sum_{mi} \frac{|D_{mi}|^2}{E_\nu^{\mathrm{TDA}} - \epsilon_m + \epsilon_i}. \tag{8.18}$$

We can solve Eq. (8.18) graphically by plotting the r.h.s. as a function of E_ν^{TDA}. We thus obtain the eigenvalues from the intersection of this function with the straight line $1/\lambda$. All solutions are sandwiched between the original shell model excitations $\epsilon_{mi} = \epsilon_m - \epsilon_i$, only one solution being pushed up (if $1/\lambda > 0$) or down (if $1/\lambda < 0$). The assumption of a finite particle-hole space seems to enter into this argumentation, but we have to remember that *ph* shell model excitations of definite parity are grouped into bunches roughly $2\hbar\omega_0$ apart. Figure 8.4 thus represents just one of these bunches.

The one excitation which is pushed down ($T=0$) or up ($T=1$) is a state formed by a *coherent* superposition of all matrix elements of the residual interaction, as can easily be shown for the degenerate case to be discussed

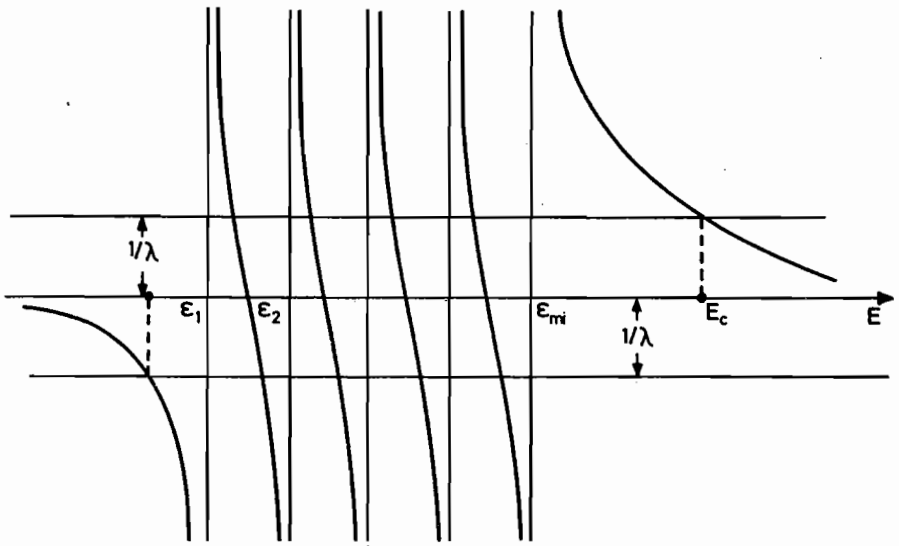


Figure 8.4. Graphical solution of Eq. (8.18).

next. By "coherent" we now mean that all matrix elements contribute with the same sign. Examples for such states are the first octupole states ($T=0$) appearing in ^{16}O, ^{40}Ca, and ^{208}Pb at low excitation energies and the giant dipole resonance ($T=1$), which we will turn to later.

8.2.2.2. Degenerate Case. If we set all ϵ_{mi} equal to ϵ, it follows from Eqs. (8.17, 8.18) that

$$C_{mi}=\left(\sum_{mi}|D_{mi}|^2\right)^{-1/2}\cdot D_{mi},$$
$$E_c^{\mathrm{TDA}}=\epsilon+\lambda\sum_{mi}|D_{mi}|^2, \tag{8.19}$$

which means that collective state is being pushed up (down) by the sum of all diagonal elements of the interaction.

This degenerate case is realized, for example, if we take into account only one major shell for particles and another major shell for holes in a spherical oscillator potential without a spin orbit term. In reality things will not be so pronounced, but the essential features are likely to be preserved.

The collectivity of the shifted state can be demonstrated for the degenerate case by studying the partitioning of the total transition probability for the different excitations. For the collective state we have

$$|\nu_c\rangle=\left(\sum_{mi}|D_{mi}|^2\right)^{-1/2}\sum_{mi}D_{mi}a_m^+a_i|\mathrm{HF}\rangle. \tag{8.20}$$

We therefore get, for the transition probability,

$$|\langle\nu_c|D|\mathrm{HF}\rangle|^2=\sum_{mi}|D_{mi}|^2, \tag{8.21}$$

a coherent superposition of the expansion coefficients (where the transition operator $D=\sum_{kk'} D_{kk'} a_k^+ a_{k'}$ shall correspond to that for the separable interaction).

For the so-called sum rule (see Sec. 8.7), we get

$$\begin{aligned}\sum_\nu |\langle \nu|D|\mathrm{HF}\rangle|^2 &= \sum_{\nu \neq \nu_c} \cdots + |\langle \nu_c|D|\mathrm{HF}\rangle|^2 \\ &= \sum_\nu \langle \mathrm{HF}|D^+|\nu\rangle\langle\nu|D|\mathrm{HF}\rangle \qquad (8.22)\\ &= \sum_{mi} \langle \mathrm{HF}|D^+|mi\rangle\langle mi|D|\mathrm{HF}\rangle = \sum_{mi} |D_{mi}|^2.\end{aligned}$$

Here we replaced the complete set $|\nu\rangle$ in the *ph* space by the complete set $|mi\rangle$. Because of (8.21), we see that the total sum rule is exhausted by the collective state

This means that in the degenerate case there is no transition probability from the ground state to any non-collective state. On the other hand, the transition probability to the collective state (8.21) is drastically enhanced. We thus have a qualitative explanation for the strong 1^- resonance shown in Fig. 8.1.

8.2.3 Particle–Particle (Hole–Hole) Tamm–Dancoff Method

As we have seen in the case of doubly magic nuclei, the simplest correlations beyond Hartree–Fock can only be taken into account by breaking the HF core and raising a nucleon from below to above the Fermi level. In moving away from doubly magic nuclei, that is, by filling nucleons into the next open shell, a quite different type of correlation may be viewed as important. Since a magic nucleus is supposed to be quite a stable entity, the correlations among the valence nucleons alone seem to be responsible for a variety of experimental facts known about these nuclei. The $s-d$ shell nuclei ranging from ^{16}O to ^{40}Ca are perhaps the most studied examples of such "open shell" nuclei. A whole theory has been developed to treat such nuclei, known most widely under the heading of *shell model configuration mixing* calculations. Many ingredients of this theory can be found in the textbook of de Shalit and Talmi [ST 63], and the most advanced calculations within this theory have been performed by the Oak Ridge group [HMW 71] and by Whitehead [Wh 72]. In this section we only want to treat the simplest possible case of such nuclei, that is, we move away from the closed shell by only two particles, thus filling in two nucleons or removing two (creating two holes). In accordance with what we have said above, and by analogy to the *ph*-case, we may therefore try the following variational ansatz (*pp-TDA method*).

$$|\tau, A+2\rangle = \sum_{m<n} C_{mn}^\tau a_m^+ a_n^+ |\mathrm{HF}\rangle. \qquad (8.23)$$

The coefficients C_{mn}^τ are supposed to be antisymmetric, that is, $C_{mn}^\tau = -C_{nm}^\tau$. In complete analogy to the *ph*-TDA case, we obtain the *pp*-TDA secular equation:

$$(E_\tau^{\mathrm{TDA}} - \epsilon_m - \epsilon_n) C_{mn}^\tau = \sum_{m'<n'} \bar{v}_{mnm'n'} C_{m'n'}^\tau. \qquad (8.24)$$

This is a linear Hermitian eigenvalue problem. The eigenvectors have to fulfill the norm and closure relations ($n < m$, $n' < m'$)

$$\sum_{m<n} C^{\tau}_{mn} C^{\tau'}_{mn} = \delta_{\tau\tau'}, \tag{8.25}$$

$$\sum_{\tau} C^{\tau}_{mn} C^{\tau *}_{m'n'} = \delta_{mm'}\delta_{nn'}. \tag{8.26}$$

As indicated, the sum in principle runs over all levels $> \epsilon_F$ and therefore includes bound and continuum states. Since this generally gives rise to matrices too big for present day computers, we usually work in a restricted subspace, taking into account one or two major shells above the Fermi level. In order to account for the levels not included, we must take a suitably "renormalized" interaction (see Sec. 4.3.2). Equation (8.24) looks just the same for the *hh*-TDA case but the particle indices (m, n) are replaced by hole indices (i, j) and the sign of the single particle energies is reversed. The solutions of the *pp*(*hh*) equation can exhibit features similar to the *ph* case. For instance, we can also get collective low-lying states, known mostly as *pairing vibrations* (see Sec. 8.3.5). Since the qualitative discussions would be very much the same as in the *ph* case, we will not go into greater detail. (For calculations in the lead region, see, for instance, [HK 72, MT 73] and references therein.)

8.3 General Considerations for Collective Modes

8.3.1 Vibrations in Quantum Mechanics

Perhaps this is the right place to interrupt our more or less formal discussion of the TDA method for a while and consider the nature of the collective states in more detail. In Chapter 1 we described states such as vibrations of a liquid drop. The word vibration may be puzzling in the quantum mechanical context, since we are in fact only dealing with stationary states. The density $\rho(\mathbf{r})$ of such a stationary state is time independent. The correspondence is roughly the same as talking about oscillations in the quantum mechanical harmonic oscillator problem. To get a time dependent density distribution $\rho(\mathbf{r}, t)$ which vibrates around the ground state density $\rho^{(0)}(\mathbf{r})$ we have to investigate a *wavepacket*

$$|\Psi(t)\rangle = |0\rangle + \sum_{\nu} c_{\nu} |\nu\rangle e^{-iE_{\nu}t/\hbar} \tag{8.27}$$

which contains the ground state $|0\rangle$ and small admixtures of excited states $|\nu\rangle$. Up to first order in the coefficients c_{ν} the corresponding density is (see Appendix D):

$$\begin{aligned} \rho(\mathbf{r}, t) &= \langle \Psi(t) | \sum_{i=1}^{A} \delta(\mathbf{r} - \mathbf{r}_i) | \Psi(t) \rangle \\ &= \rho^{(0)}(\mathbf{r}) + \delta\rho(\mathbf{r}, t) \end{aligned} \tag{8.28}$$

with

$$\delta\rho(\mathbf{r}, t) = \sum_{\nu} c_{\nu} \langle 0 | \sum_{i=1}^{A} \delta(\mathbf{r} - \mathbf{r}_i) | \nu \rangle e^{-iE_{\nu}t/\hbar} + \text{c.c.} \tag{8.29}$$

A Fourier analysis of this density gives the contributions of the different excited states

$$\rho^{(1)\nu}(\mathbf{r}) = \langle 0| \sum_{i=1}^{A} \delta(\mathbf{r}-\mathbf{r}_i)|\nu\rangle. \tag{8.30}$$

These quantities are called *transition densities*.

In general, we expect oscillations not only in the local part of the density as given in Eq. (8.28), but also in the nonlocal part, that is, we are interested in the transition density matrix. As discussed in Appendix D, in a shell model basis, it has the form

$$\rho_{pq}^{(1)\nu} = \langle 0|a_q^+ a_p|\nu\rangle. \tag{8.31}$$

In the TDA approach these matrix elements are given by the coefficients C_{mi}^{ν} in Eq. (8.5), that is, in TDA we have

$$\rho_{mi}^{(1)\nu} = C_{mi}^{\nu}, \qquad \rho_{ii}^{(1)\nu} = \rho_{mm}^{(1)\nu} = \rho_{im}^{(1)\nu} = 0. \tag{8.32}$$

There are many different types of modes a nucleus can assume according to its various degrees of freedom: A nucleus can be deformed, displaced, or compressed; the densities of the protons and neutrons can vibrate *in* and *out* of phase; there can be vibrations in spin and isospin; we have pairing modes and many more things of which only a few elementary features will be discussed here (see also Chap. 13).

8.3.2 Classification of Collective Modes

Since we are dealing with harmonic vibrations in this chapter, in the following we want to investigate their qualitative properties in more detail. To see the basic structures we restrict our discussion to spherical closed shell nuclei. The best suited example is ^{208}Pb. These considerations do, however, also apply for the other regions in the periodic table. In fact, as we saw in the introduction, many of these collective excitations show rather classical properties which depend smoothly on the mass number.

Collective *ph*-vibrations can be characterized by the properties of the corresponding pair operators [Mo 76a+b]

$$(a^+a). \tag{8.33}$$

There are a number of exact symmetries of the Hamiltonian such as rotational invariance, parity conservation, and charge conservation with the corresponding quantum numbers I, π and T_z. In addition, we have (as discussed in Sec. 2.6.3) the approximate isospin symmetry for light and heavy nuclei with the quantum number T. Since we start with a spherical closed shell nucleus in its ground state ($I^{\pi}=0^+$), the angular momentum and the parity of the excitation are determined by the corresponding quantum numbers of the coupled pair $(a^+a)_{I^{\pi}}$. The same also holds true in light ($N=Z$) nuclei (with $T=0$ in the ground state) for the isospin. For

heavy nuclei, the isospin of the ground state is $T_0=(N-Z)/2$ and it has to be coupled to the isospin of the *ph* pair to the total isospin $T=T_0$ or $T=T_0+1$.* We usually classify the excitation by the numbers ΔT of the *ph* pairs a^+a. Therefore, we have $\Delta T=0$ or $\Delta T=1$ excitations, and a $\Delta T=1$ excitation does not necessarily mean $T=T_0+1$.

In addition to these general considerations about symmetries, which give, for instance, information about the angular distributions, we want to classify the collective motion more precisely according to the different degrees of freedom in the *ph* pair

$$a^+(\mathbf{r}, s, t)a(\mathbf{r}', s', t'). \tag{8.34}$$

(a) In the first place, there are vibrations of the *local* density $\rho(\mathbf{r})$ of the nucleus in space. Since the angular dependence of such vibrations is already completely determined by the angular momentum, we only can allow for different *radial shapes*. The simplest examples in this context are the surface vibrations of a sphere with a sharp surface, which we have discussed in Section 1.4 in great detail. From Eq. (1.7) we find that the transition density (8.30) for such a surface vibration with a sharp surface is given by

$$\rho^{(1)}(r)=\begin{cases} \pm\rho_0 & \text{for } r \text{ between } R_0 \text{ and } R(\vartheta,\varphi), \\ 0 & \text{elsewhere,} \end{cases} \tag{8.35}$$

that is, the density in the nuclear interior stays constant. The entire transition is concentrated in the surface region.

We can also imagine other radial dependences of $\rho^{(1)}$, where the density in the interior is also changed. Such excitations are called *breathing modes*.

(b) A quite different type of motion, which also involves only the spatial degrees of freedom is given by vibrations in the nonlocal part of the transition density matrix $\rho^{(1)}(\mathbf{r}, \mathbf{r}')$. An example is the *nuclear twist* mode, where the local density of the nucleus stays unchanged and only the intrinsic velocity distribution oscillates (see [HE 77]).

(c) The isospin is an additional degree of freedom. There we have to distinguish between excitations in the same nucleus $\Delta T_z=0$ and isospin flip processes $\Delta T_z=-1$, which belong to the neighboring nucleus with the charge $Z+1$ and the neutron number $N-1$. In the first case, the *ph* pair can be coupled to either $\Delta T=0$ or to $\Delta T=1$. For all the modes discussed so far, we therefore have two options: The protons can vibrate in phase ($\Delta T=0$) or out of phase ($\Delta T=1$) with the neutrons. At a fixed point $\mathbf{r}$ we then eventually have an oscillating charge distribution. This corresponds to *polarization waves* in the nucleus. A case of vibrations with $\Delta T_z=-1$ is the analog resonances, which will be discussed in Section 8.3.4.

* $T=T_0-1$ is only possible in proton particle–neutron hole excitations that correspond to states in the neighboring odd–odd nucleus.

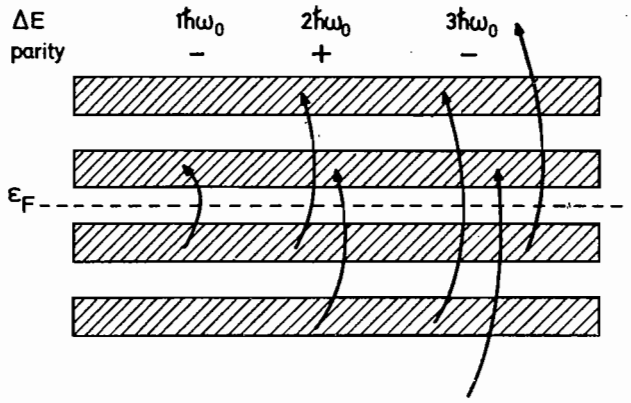


Figure 8.5. Schematic picture of the shell structure in a closed shell nucleus and possible $1p-1h$ excitations.

(d) Finally, we also have the *spin* degree of freedom. The corresponding vibrations produce spin waves.

The classification of different modes given so far was, to a large extent, based on considerations within the coordinate space and emphasized its classical aspects. In finite realistic nuclei however, these modes are not precisely realized. There are always more or less drastic admixtures of less collective states. Therefore, it is often very useful to apply quite a different classification, based on the *shell model* in its simplest version—the harmonic oscillator.

Figure 8.5 shows schematically the shell structure for a closed shell nucleus (as ^{208}Pb). We have alternating shells with positive and negative parities separated by an energy distance of roughly $\hbar\omega_0$. For a closed shell nucleus, the Fermi surface lies just between two shells and we therefore find *ph*-pairs of a $1\hbar\omega_0$ excitation with negative parity, or a $2\hbar\omega_0$ excitation with positive parity, or a $3\hbar\omega_0$ excitation with negative parity, and so on. For nuclei between closed shells we also have, in addition, $0\hbar\omega_0$ excitations within the shell.

In realistic nuclei, the degenerate oscillator shells split up and a few levels with the wrong parity are shifted down into a lower shell by the **ls** term. This also causes $1\hbar\omega_0$ excitations with positive parity for closed shell nuclei such as ^{208}Pb. But the basic features of this pictures also hold in realistic cases.

Although we have superpositions of many *ph* pairs in the collective excitation and drastic energy shifts due to the correlations as discussed in Section 8.2.2.2, this gross structure is conserved to a large extent.* For $J^\pi = 3^-$, for instance, we find a very collective state consisting mainly of $1\hbar\omega_0$ excitations and a second one consisting mainly of $3\hbar\omega_0$ excitations [MNS 76, St 79].

*One can also show this analytically in a schematic model with several bunches of degenerate levels in analogy to Sec. 8.2.2.2.

It is evident that this shell structure has a great influence on the excitation energy of the low-lying collective states (shown schematically in Fig. 1.6), because they depend closely on the *ph* energy of the lowest *ph* pair. In fact, here we observe drastic shell effects (see Fig. 1.7). For the high-lying states, the giant resonances, this is no longer the case. The energy distance between two major shells varies only smoothly over the periodic table [$\hbar\omega_0 \propto A^{-1/3}$; see Eq. (2.12)]. We therefore expect a similar smooth A-dependence of the resonance energy for giant resonances.

8.3.3 Discussion of Some Collective *ph*-Vibrations

In a sense the most trivial collective motion is a *translation* of the whole nucleus, generated by the linear momentum **P**. Since it commutes with the two-body Hamiltonian H, the exact ground state $|0\rangle$ is an eigenfunction of **P**. However, if we use an approximation $|0\rangle_{\text{app}}$ for the ground state which violates this symmetry, $\mathbf{P}|0\rangle_{\text{app}}$ are new states. They look like collective excitations. We call them *spurious*, because they do not correspond to physical excitations. Within the angular momentum coupling scheme, their quantum numbers are $J=1^-$, $T=0$ (**P** is a vector operator that does not depend on the isospin). They correspond to a translation of the whole nucleus. There is no restoring force against this motion and therefore they should have zero excitation energy

$$H\mathbf{P}|0\rangle_{\text{app}} = \mathbf{P}H|0\rangle_{\text{app}} \simeq E_0\mathbf{P}|0\rangle_{\text{app}}. \tag{8.36}$$

However, this depends on the approximation. For instance, in the TDA, $\mathbf{P}|\text{HF}\rangle$ is a superposition of particle–hole states, which is generally not an eigenstate of the TDA equations (8.10). Instead, it is a mixture of all the TDA eigenstates, which form a complete set in the $1p-1h$ space. Since its energy is nevertheless close to E_0 (Eq. 8.36), it has large components of the low-lying TDA states. This means some low-lying solutions of the TDA equation (8.10) have large spurious contributions. As long as we are working in a spherical basis with good spin and isospin, only the $J=1^-$, $T=0$ states are influenced. They have to be treated separately in order to extract the spurious components. For details, see Chapter 11.

The best known and the most thoroughly investigated giant resonance is the *giant dipole state* (see for instance [Sp 68]). It is a several MeV broad resonance, which has been observed over the whole periodic table at an energy of roughly

$$E_{1^-} \propto 80A^{-1/3 \div -1/6}\,[\text{MeV}]. \tag{8.37}$$

A typical case is shown in Fig. 8.1. The experimental spectrum usually shows some fine structure. It has its origins in the specific properties of nuclei and is not constant over the periodic table. Particular *ph*-components may show up—as in ^{16}O [EF 57]—or shell effects may cause a splitting of the resonance. For instance, passing from spherical to de-

formed nuclei, we observe two kinds of dipole vibrations, one parallel to the symmetry axis and one orthogonal to it. The splitting of the energies allows an experimental determination of the deformation parameter [Da 58, Ok 58, BBC 71].

The giant dipole resonance is excited, for instance, by photo-absorption of γ-radiation. In Section B.7 we derive the cross section for the excitation of a final state $|\nu\rangle$ by a photon of energy E

$$\sigma_\nu(E)=\frac{4\pi^2e^2}{\hbar c}(E_\nu-E_0)|\langle\nu|D|0\rangle|^2\delta(E-E_\nu+E_0). \tag{8.38}$$

D is the dipole operator (B.88) for $E1$-radiation in the z-direction.

$$D=\frac{NZ}{A}\left(\frac{1}{Z}\sum_{p=1}^{Z}z_p-\frac{1}{N}\sum_{n=1}^{N}z_n\right)=-\sum_{i=1}^{A}t_3^{(i)}(\mathbf{r}_i-\mathbf{R})_z. \tag{8.39}$$

It is proportional to the z-component of the distance between the two centers of gravity for protons and neutrons.

The total cross section for dipole absorption σ_{total} is obtained from (8.38) by summing over all final states $|\nu\rangle$ and integrating over the energy:

$$\sigma_{\text{total}}=\sum_\nu\int_0^\infty\sigma_\nu(E)\,dE=\frac{4\pi^2e^2}{\hbar c}\sum_\nu(E_\nu-E_0)|\langle\nu|D|0\rangle|^2. \tag{8.40}$$

It is proportional to the energy-weighted dipole sum rule (see also Sec. 8.7)

$$S_1(D)=\sum_\nu(E_\nu-E_0)|\langle\nu|D|0\rangle|^2 \tag{8.41}$$

and can be evaluated in closed form:

$$S_1(D)=\tfrac{1}{2}\langle0|[D,[H,D]]|0\rangle,$$

where H is the exact two-body Hamiltonian and $|0\rangle$ is the exact ground state. If we assume that the two-body potential has no velocity dependence and no exchange mixtures (Wigner force), we easily calculate in analogy to (8.158):

$$\langle0|[D,[H,D]]|0\rangle=\frac{NZ}{A}\frac{\hbar^2}{m}. \tag{8.42}$$

For the dipole sum we therefore have the Thomas–Reiche–Kuhn sum rule:

$$\sigma_{\text{total}}=\int_0^\infty\sigma(E)\,dE=\frac{2\pi^2e^2\hbar}{mc}\frac{NZ}{A}\simeq0.06\frac{NZ}{A}\,[\text{MeV}\cdot\text{barn}]. \tag{8.43}$$

This value is enlarged by 40–80% if the two-body interaction contains exchange mixtures (see, e.g., Brenig [Br 65b]) and tensor forces [WKB 73], which then seems to yield roughly the right magnitude of the overall energy integrated cross section. It thus turns out that in ^{16}O approximately 3% of the dipole sum lies in low-lying 1^- states, which are almost pure shell model $p-h$ resonances below 20 MeV, whereas the giant resonance peak contains about 50%, clearly reflecting its collective character. The rest of the dipole sum is to be found beyond 30 MeV [ABC 63, FH 62a, b, Ha 65b, BS 64, ADW 78]. For heavy nuclei the giant dipole resonance can

exhaust up to 100% of the sum rule. The experiment therefore shows that a very large part of the dipole sum (8.43) is exhausted by the giant dipole resonance, and this means that this resonance has a large overlap with the state

$$|D\rangle = D|0\rangle = \frac{NZ}{A}\left\{\frac{1}{Z}\sum_{pp'}\langle p|z|p'\rangle a_p^+ a_{p'} - \frac{1}{N}\sum_{nn'}\langle n|z|n'\rangle a_n^+ a_{n'}\right\}|0\rangle, \tag{8.44}$$

which has become known as the Goldhaber–Teller state [GT 48] (see Sec. 13.3.3).

The quantum numbers of the state are identical with those of the giant resonance $I^\pi = 1^-$. In the case of the isospin, we have to distinguish two cases.

(a) For light $N = Z$ nuclei the Coulomb force can be neglected. The ground state $|0\rangle$ then has $T = 0$. The operator D is a vector in isospin space, as we see from Eq. (B.88). Angular momentum coupling rules therefore show that $D|0\rangle$ has $T = 1$.

(b) For heavy nuclei the isospin is again conserved. The ground state has $T_0 = (N - Z)/2$. Using similar arguments as in Sec. 2.6.3 we can convince ourselves that the dipole state $|D\rangle$ has the same isospin T_0 by showing that

$$T_+|D\rangle \simeq 0. \tag{8.45}$$

This does not, however, mean that there are no other 1^- states with $T = T_0 + 1$ in the excitation spectrum of the nucleus. We simply do not excite them by dipole γ-rays.

The structure of the operator D shows that the giant dipole resonance consists mainly of $1\hbar\omega_0$ excitations. Because of the repulsive character of the residual interaction it is shifted to higher energies.

Besides this well known dipole state, in the last ten years a number of other giant resonances of higher multipolarity have been found. The *giant quadrupole resonance* is especially well established over the whole periodic table at an energy somewhat less than the giant dipole resonance [PW 71, FT 72, LB 72, HGA 74, Sa 74, KWB 75, YMR 76, Be 76]

$$E_{2^+} \simeq 60 - 65 A^{-1/3}\ [\text{MeV}]. \tag{8.46}$$

It exhausts 40–100% of the energy weighted sum rule of the isoscalar quadrupole operator*

$$r^2 Y_{20}. \tag{8.47}$$

* Unfortunately, in a direct comparison with experiment we cannot use photo-absorption, as in the dipole case, because (i) the photon field does not contain the isovector part for $\lambda > 1$, and (ii) in practice the photonuclear reactions proceed primarily through the dipole mode. The most important experimental tool used to excite such vibrations is inelastic scattering of electrons and heavier particles such as protons, deuterons, and alpha particles [Sa 76]. The corresponding excitation operators have a more complicated structure (see, for instance, [Ue 71, Sa 72a+b, 74, HMS 75] and the sum rules are no longer model independent. Therefore, we will not go into the details of the analysis of such experiments.

It consists, as numerical calculations show [RS 74a], mainly of $2\hbar\omega_0$ excitations, and is shifted downwards in energy by the residual interaction to around $1.4\hbar\omega_0$.* (The corresponding isovector mode is expected at $3.3\hbar\omega_0$ [Ha 72].) The rest of the isoscalar quadrupole strength lies in the low-lying collective 2^+ state, which shows drastic shell effects (Fig. 1.7) and consists mainly of $0\hbar\omega_0$ excitations.

Recently, a *breathing mode* was also observed [MMW 75, ST 77, HBI 77, YRM 77, BBL 80], a spherical density oscillation with the quantum number $J^\pi = 0^+$, $T = 0$. As we shall see in Fig. 8.14, its transition density is not only concentrated on the nuclear surface but involves changes of the nuclear density over the whole volume. Its excitation energy lies at

$$E_{0^+} \simeq 80 A^{-1/3} \text{ MeV}. \tag{8.48}$$

This energy is a direct measure of the nuclear incompressibility (5.95) [Wa 62a, BGG 76]. In ^{208}Pb the experimental breathing mode lies at 13.8 MeV, a value corresponding to $K \simeq 200$ MeV, which is in good agreement with theoretical predictions [RS 74a, SZR 74, WMR 77].

* In deformed regions we observe a splitting of the giant quadrupole resonance into three components with $K = 0$, 1 and 2 [KMY 75].

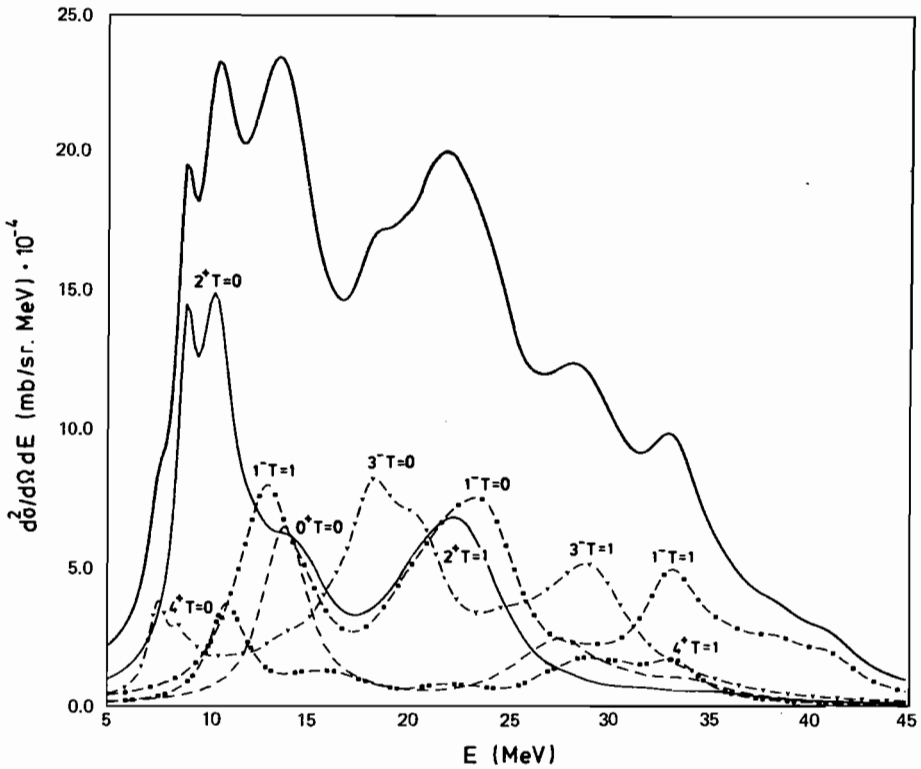


Figure 8.6. The electron spectrum for 90 MeV (e, e') at 75° calculated with microscopic RPA wave functions. The numbers in the figure denote the angular momentum parity and isospin of the corresponding cross-section contribution. The thick line is the sum of all the different contributions. (From [WKS 78].)

Magnetic resonances are excited by operators which involve the spin and the orbital angular momentum operators. They have the quantum numbers 1^+, 2^-, 3^+,... and have been predicted by several calculations (see, for instance, [RS 74a, SWK 76, FRS 78]). Their experimental evidence, however, is still an object of discussion.

Further giant resonances in the spectra have been observed in various experiments (see, for instance, [PBD 74, TIS 75, Pa 75a, SEB 74]), but the spin and parity assignments are not completely established. As an illustrative picture we show in Fig. 8.6 a calculation of the (e, e') spectrum for the giant resonance region of ^{208}Pb. We see clearly that the total spectrum can only be obtained by a complicated superposition of rather different resonances.

8.3.4 Analog Resonances

Nuclear reactions for heavy nuclei ($N > Z$), like, for example, ^{208}Pb (p, n) ^{208}Bi, exhibit strong collective resonances called *analog states* for reasons we will learn about in a moment.*

To explain the structure of these states we use the fact that the nuclear wave functions in heavy nuclei are, to a rather good approximation, eigenstates of the isospin operator $\mathbf{T}^2$ (see Sec. 2.6.3). The ground state and the low-lying excited states have the quantum numbers $T = T_3 = (N-Z)/2$.

Starting from such a state $|I, T, T_3 = T\rangle$ at an energy E, we can define its *analog state* by the application of the isospin lowering operator T_-.

$$T_-|I, T, T_3 = T\rangle = \sqrt{2T}\,|I, T, T_3 = T-1\rangle. \tag{8.49}$$

It belongs to the "daughter" nucleus $(N-1, Z+1)$ and is again an eigenstate of H with the energy

$$E_a = E + \Delta_c. \tag{8.50}$$

To see this we decompose the exact Hamiltonian H into the nuclear part H_{nucl} which commutes with T_- and the Coulomb interaction V_c

$$HT_- = (H_{\text{nucl}} + V_c)T_- = T_-H + [V_c, T_-]. \tag{8.51}$$

The commutator $[V_c, T_-]$ is of the form

$$\begin{aligned}[V_c, T_-] &= \sum_{i<j} V_c(i, j)\left[\left(\tfrac{1}{2} - t_3^{(i)}\right)\left(\tfrac{1}{2} - t_3^{(j)}\right), T_-\right] \\ &= \sum_i V_c(i) t_-^{(i)}\end{aligned} \tag{8.52}$$

with the average Coulomb field

$$V_c(i) = \sum_{j \neq i} V_c(i, j)\left(\tfrac{1}{2} - t_3^{(j)}\right). \tag{8.53}$$

Under the assumption that this single-particle potential in the interior of the nucleus has a constant value of roughly Δ_c (see Fig. 2.7), we obtain for each eigenstate $|n\rangle$ of the "parent" nucleus (N, Z) an analog state $T_-|n\rangle$ in the

* For the literature, see, for instance, [AHK 72].

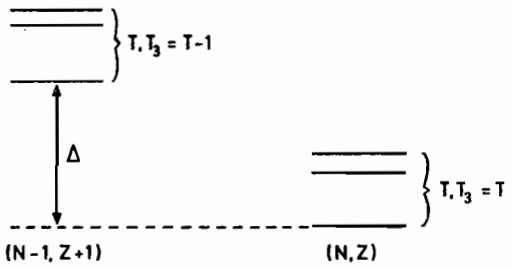


Figure 8.7. Schematic representation of the spectrum of analog states.

"daughter" nucleus $(N-1, Z+1)$

$$HT_-|n\rangle = E_n T_-|n\rangle + [H, T_-]|n\rangle \simeq (E_n + \Delta_c) T_-|n\rangle. \tag{8.54}$$

This means that in the daughter nucleus there is a sequence of states with the same energy spacings as the low-lying levels of the parent nucleus itself, having isospin

$$T = \frac{N-Z}{2} = \frac{N-Z-2}{2} + 1, \tag{8.55}$$

which is one unity larger than the isospin of the daughter nucleus ground state (Fig. 8.7).

In order to learn more about the structure of the analog resonances we will forget for the moment the residual interaction used in the description of the ground state of the parent nucleus (N, Z). This will be given by a pure shell model description, represented graphically in Fig. 8.8.

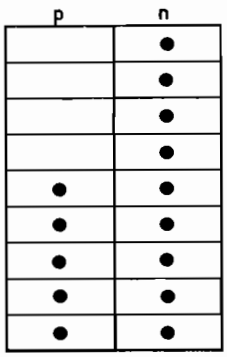
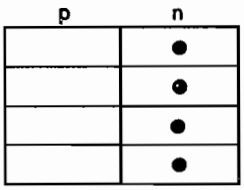


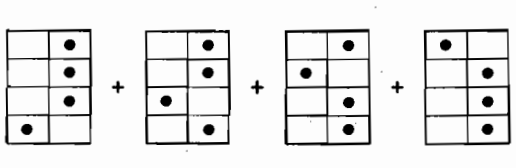
Figure 8.8. Schematic representation of a pure shell model determinant.

We assume that the protons and neutrons, which occupy the states with the same quantum numbers (besides t_3, of course) form an inert core and are coupled to $T=0$. In the following, we shall neglect this inert core, so that the ground state of the mother nucleus is given by

$|I, T, T_3 = T\rangle =$  . (8.56)

With $T_- = \sum_i t_-^{(i)}$, we obtain for the analog state of the daughter nucleus:

$$|I, T, T-1\rangle = \frac{1}{\sqrt{2T}} \left\{ \right.$$



$$\left. \right\}$$

$$= \frac{1}{\sqrt{2T}} \sum_\mu |\mu\rangle. \tag{8.57}$$

The analog state in Eq. (8.57) is a coherent superposition of certain *ph*-states formed by replacing, in (8.56), the neutrons by protons one by one in the corresponding states.

There are, of course, other linear combinations of the states $|\mu\rangle$ orthogonal to the analog state $|I, T, T_3 = T-1\rangle$:

$$\sum_\mu c_\mu |\mu\rangle \qquad \text{with } \sum_\mu c_\mu = 0.$$

The operator T_+ annihilates them

$$T_+ \sum_\mu c_\mu |\mu\rangle \propto \left(\sum_\mu c_\mu \right) \cdot |I, T, T_3 = T\rangle = 0. \tag{8.58}$$

Therefore, they must have isospin $T-1$, which is the isospin of the ground state of the daughter nucleus. Without residual interaction they lie at the same energy as the analog state and have equal spin and parity.

Experimentally, the analog states are observed as very sharp resonances at high energies [AW 61]. Their width is so small, because the isospin selection rule forbids a rapid decay into the other states which form a continuous background.

To see how they can be treated with the TDA, we start with a HF potential which is the same for protons and neutrons. Only the energy levels of protons and neutrons are shifted with respect to one another by an amount $\Delta_\epsilon = \Delta_c - \Delta_s$ [see Eq. (2.27)]. The quantity Δ_s has its origin in the symmetry energy. In our approximation this is the only difference in the self-consistent potential of protons and neutrons resulting from the nuclear force.

If we diagonalize the residual interaction in the space of the states $|\mu\rangle$ we obtain a splitting between the analog resonance and the other states. Assuming a separable force we end up with the degenerate model of Section 8.2.2.2. Again, it concentrates all strength in the analog state and shifts its energy from the unperturbed value $\Delta_\epsilon = \Delta_c - \Delta_s$ to its proper value Δ_c, because it is exactly the proton–neutron interaction which causes the symmetry energy Δ_s. For details, see the textbook of G. E. Brown [Br 64, (second edition 1967)] and other references cited in the review article [AHK 72].

8.3.5 Pairing Vibrations

As indicated in the treatment of the *pp*(*hh*)–TDA method, there is a great formal similarity between the *ph* and *pp*(*hh*) cases. In fact, it also turns out that there are collective *pp*(*hh*) solutions of the corresponding TDA equation (8.24). A good example is the 0^+ ground state of ^{206}Pb. The amplitudes C_{mn}^τ according to Eq. (8.23) enter directly into the expression of, for

example, a ^{208}Pb (p, t) ^{206}Pb pickup or a ^{208}Pb (t, p) ^{210}Pb stripping cross section [Gl 65]. The fact that the 0^+ ground state of ^{206}Pb is populated by such a reaction an order of magnitude larger than the other 0^+ states of ^{206}Pb, clearly indicates the coherent contribution of the $C_{ij}^{0^+}$ amplitudes and therefore the collectivity of this state. Similar to this is the situation for ^{210}Pb. These rather strongly bound entities of two particles in (0^+) states is due to the short-range part of the interaction and has already been discussed at the beginning of Chapter 6. Two particles form such a stable entity that they can be multiplied, added to, or removed from a nucleus (like, for example, ^{208}Pb) almost like independent particles (which behave very much like bosons because of their integer spin). The spectrum should therefore be approximately harmonic which, in fact, is the case for the lead isotopes (Fig. 8.9) [BM 75]. This harmonic spectrum is what has been termed the spectrum of "pairing vibrations." We can also produce excited 0^+ pairing vibrations in these nuclei. For example, let us remove from ^{208}Pb a 0^+ pairing phonon, which leads us to ^{206}Pb. At the same time, let us add a phonon leading from the ^{208}Pb to the ^{210}Pb ground state; this clearly gives an excitation in ^{208}Pb $2\hbar\Omega$ apart and indeed such a strongly excited state in ^{210}Pb (p, t) ^{208}Pb and ^{206}Pb (t, p) ^{208}Pb reactions is very well known and lies at 4.9 MeV in ^{208}Pb.

The fact that particle–particle correlations lead to collective nuclear modes—now usually called pairing vibrations—was recognized by H. Schmidt [Sch 64a, b] and Bohr [Bo 64] and worked out in much detail later on by Bès and Broglia [BB 66] (for a review see [BHR 73] and the references therein).

It is a quite general concept which describes the pairing correlations on the same level as the deformations. Starting from a spherical closed shell nucleus, we can excite *quadrupole vibrations*, which have an *angular momentum* different from the ground state, and *pairing vibrations*, which belong to nuclei with a different *particle number*. Ground state correlations induced by such vibrations (see Sec. 8.4.6) give in the first case a virtual quadrupole deformation and in the second case a virtual excitation of collective $2p-2h$ states. As we go away from a closed shell, the nuclei get softer, that is, the correlations get larger. Finally we get a phase transition to deformed states—induced by the *ph*-correlations; or to superfluid states—induced by

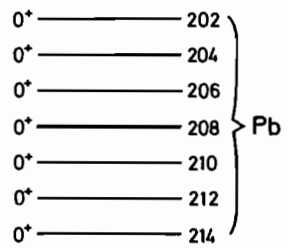


Figure 8.9. Schematic representation of the ground state energies in the even Pb isotopes.

the *pp*-correlations. In such a case, we have a condensate of quadrupole or pairing bosons (see Sec. 8.8.2).

We know that we have surface vibrations in deformed nuclei. These are oscillations around a stable deformation. In the same way, we can have pairing vibrations in superfluid nuclei. They correspond to oscillations of the gap parameters around their ground state values.

8.4 Particle–Hole Theory with Ground State Correlations (RPA)

One feature of our approximate ansatz (8.5) was the fact that we could build correlations into the excited states; the ground state however, remains unchanged in the TDA method. This complete neglect of the residual interaction V_R in the ground state certainly influences the results. To remove this drawback of the TDA method, we could think of retaining also the $(2p-2h)$ components in Eq. (8.2). This is not practicable, since the corresponding matrices would become prohibitively large. One way out is a generalization of the TDA method in which we take, instead of the HF ground state, one in which a certain class of correlations has been summed. Although the discussion of the problem stays more or less the same as for the TDA method, we will have to pay a certain price for this refinement of our theory. Our new equations will no longer follow out of a variational principle, and thus we can generally get imaginary solutions. We will derive the new equations [random-phase approximation (RPA)]* in close analogy to the TDA method using a different technique which will suggest quite naturally the necessary generalizations leading to the RPA.

8.4.1 Derivation of the RPA Equations

To get a better understanding of what is done in the random phase approximation, we will first re-derive the TDA equations using a different technique, the so-called equation of motion method [Ro 68a, 70]. We start with a set of exact eigenstates of the Hamiltonian H

$$H|\nu\rangle = E_\nu|\nu\rangle. \tag{8.59}$$

It is possible to define operators† Q_ν^+ and Q_ν in such a way that

$$|\nu\rangle = Q_\nu^+|0\rangle \qquad \text{and} \qquad Q_\nu|0\rangle = 0. \tag{8.60}$$

*This approximation has been introduced by Bohm and Pines [BP 53] in the theory of plasma oscillations. Other derivations have been given by several authors [GB 57, Hu 57, Sa 57, An 58].

†The frequent use later on of the letter Q in the sense of a coordinate should not be confused with its present meaning.

Q_ν^+, for example, can be chosen as

$$Q_\nu^+ = |\nu\rangle\langle 0|.$$

From the Schrödinger equation (8.59) we get the equation of motion

$$[H, Q_\nu^+]|0\rangle = (E_\nu - E_0)Q_\nu^+|0\rangle. \tag{8.61}$$

Multiplying from the left with an arbitrary state of the form $\langle 0|\delta Q$ we get*

$$\langle 0|[\delta Q, [H, Q_\nu^+]]|0\rangle = (E_\nu - E_0)\langle 0|[\delta Q, Q_\nu^+]|0\rangle. \tag{8.62}$$

We can use the commutator, because $\langle 0|Q_\nu^+ = \langle 0|HQ_\nu^+ = 0$. Until now we were exact and, since the variation of $\delta Q|0\rangle$ exhausts the whole Hilbert space, (8.62) corresponds to the full Schrödinger equation (8.59). First we re-derive the TDA equation (8.10) by approximating the exact ground state $|0\rangle$ by the HF state $|\mathrm{HF}\rangle$ and the operator Q_ν by the collective *ph*-operator

$$Q_\nu^+ = \sum_{mi} C_{mi}^\nu a_m^+ a_i. \tag{8.63}$$

By this approximation, we restrict ourselves to the space of $1p-1h$ excitations, that is, we set, $\delta Q|0\rangle = \sum_{mi} a_m^+ a_i|\mathrm{HF}\rangle\delta C_{mi}$ and from (8.62) get

$$\sum_{nj}\langle \mathrm{HF}|[a_i^+ a_m, [H, a_n^+ a_j]]|\mathrm{HF}\rangle C_{nj}^\nu = E_\nu^{\mathrm{TDA}} C_{mi}^\nu, \tag{8.64}$$

where E_ν^{TDA} is the excitation energy in TDA approximation. This is exactly the TDA equation we derived earlier (8.10). In deriving this equation we have not used a variational principle this time. The above procedure has, however, the advantage that it can be generalized in a straightforward way. There is no reason why we should not use, in Eq. (8.63), a more general vibration creation operator which then also implies, as we shall see, a more general ground state. If we think of a ground state containing $2p-2h$ correlations, as indicated by Eq. (8.2), we can not only create a *ph* pair but also destroy one. The most straightforward generalization of Eq. (8.63) is therefore:

$$Q_\nu^+ = \sum_{mi} X_{mi}^\nu a_m^+ a_i - \sum_{mi} Y_{mi}^\nu a_i^+ a_m, \tag{8.65}$$

where the minus sign has been chosen for convenience. The RPA ground state $|\mathrm{RPA}\rangle$ is defined by analogy to (8.60) by†

$$Q_\nu|\mathrm{RPA}\rangle = 0. \tag{8.66}$$

We will later on deduce from this condition an explicit expression for the ground state. Instead of only one matrix C_{mi}^ν we now have two matrices X_{mi}^ν and Y_{mi}^ν. We also have two kinds of variations $\delta Q|0\rangle$, namely $a_m^+ a_i|0\rangle$ and

* If we express Q by the operators $a_p^+ a_q$, $a_p^+ a_q^+ a_r a_s$ with coefficients C_{pq}, and C_{pqrs}, then δQ is given by $\partial Q/\partial C \cdot \delta C$ for arbitrary variations δC.

† One usually assumes $|\mathrm{RPA}\rangle$ to be the ground state of an even system with closed shells ($I^\pi = 0^+$). Rowe et al. also investigated the possibility of constructing an RPA based on an open shell configuration (*open shell* RPA; see [RW 69, 70, NR 71, RN 75]).

$a_i^+ a_m|0\rangle$. Therefore, from (8.62) we get two sets of equations:

$$\langle \mathrm{RPA}|\left[a_i^+ a_m, \left[H, Q_\nu^+\right]\right]|\mathrm{RPA}\rangle = \hbar\Omega_\nu \langle \mathrm{RPA}|\left[a_i^+ a_m, Q_\nu^+\right]|\mathrm{RPA}\rangle,$$
$$\langle \mathrm{RPA}|\left[a_m^+ a_i, \left[H, Q_\nu^+\right]\right]|\mathrm{RPA}\rangle = \hbar\Omega_\nu \langle \mathrm{RPA}|\left[a_m^+ a_i, Q_\nu^+\right]|\mathrm{RPA}\rangle, \quad (8.67)$$

where $\hbar\Omega_\nu$ is the excitation energy of the state $|\nu\rangle$. These equations contain only expectation values of four Fermion operators, which are still very complicated to calculate, because we do not as yet know the ground state $|\mathrm{RPA}\rangle$.

We content ourselves with an approximation usually known as the "*quasi-boson approximation*" [BET 61, Br 64]. If we assume that the correlated ground state does not differ very much from the HF ground state, we can calculate all expectation values in the HF approximation, for example,

$$\begin{aligned}\langle \mathrm{RPA}|\left[a_i^+ a_m, a_n^+ a_j\right]|\mathrm{RPA}\rangle &= \delta_{ij}\delta_{mn} - \delta_{mn}\langle \mathrm{RPA}|a_j a_i^+|\mathrm{RPA}\rangle \\ &\quad - \delta_{ij}\langle \mathrm{RPA}|a_n^+ a_m|\mathrm{RPA}\rangle \\ &\simeq \langle \mathrm{HF}|\left[a_i^+ a_m, a_n^+ a_j\right]|\mathrm{HF}\rangle = \delta_{ij}\delta_{mn}. \quad (8.68)\end{aligned}$$

The name "quasi-boson" approximation comes from the fact that Eq. (8.68) would be an exact relation if the *ph* creation and annihilation operators obeyed the commutation relations for boson field operators. Equation (8.68), however, violates the Pauli principle because we have neglected terms coming from the commutator. The quality of this approximation can only be checked from realistic calculations (see also Chap. 9).

Within the quasi-boson approximation, the amplitudes X_{mi}^ν and Y_{mi}^ν have a very direct meaning: their absolute squares give the probability of finding the states $a_m^+ a_i|0\rangle$ and $a_i^+ a_m|0\rangle$ in the excited state $|\nu\rangle$, that is, the *ph* and *hp* matrix elements of the transition density $\rho^{(1)}$ of equation (8.31):

$$\begin{aligned}\rho_{mi}^{(1)\nu} &= \langle 0|a_i^+ a_m|\nu\rangle \simeq \langle \mathrm{HF}|\left[a_i^+ a_m, Q_\nu^+\right]|\mathrm{HF}\rangle = X_{mi}^\nu, \\ \rho_{im}^{(1)\nu} &= \langle 0|a_m^+ a_i|\nu\rangle \simeq \langle \mathrm{HF}|\left[a_m^+ a_i, Q_\nu^+\right]|\mathrm{HF}\rangle = Y_{mi}^\nu. \quad (8.69)\end{aligned}$$

Equations (8.67) can now be written in a very compact form:

$$\begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix}\begin{pmatrix} X^\nu \\ Y^\nu \end{pmatrix} = \hbar\Omega_\nu \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} X^\nu \\ Y^\nu \end{pmatrix} \quad (8.70)$$

with $(X^\nu)_{mi} = X_{mi}^\nu$; $(Y^\nu)_{mi} = Y_{mi}^\nu$; and

$$\begin{aligned}A_{minj} &= \langle \mathrm{HF}|\left[a_i^+ a_m\left[H, a_n^+ a_j\right]\right]|\mathrm{HF}\rangle = (\epsilon_m - \epsilon_i)\delta_{mn}\delta_{ij} + \bar{v}_{mjin}, \\ B_{minj} &= -\langle \mathrm{HF}|\left[a_i^+ a_m\left[H, a_j^+ a_n\right]\right]|\mathrm{HF}\rangle = \bar{v}_{mnij}. \quad (8.71)\end{aligned}$$

The matrix A is Hermitian and the matrix B is symmetric.

Equation (8.70), together with (8.71), is called the *RPA equation*. We get back the TDA equation by putting all Y_{mi}^ν equal to zero. They are therefore a measure for the correlations in the ground state.

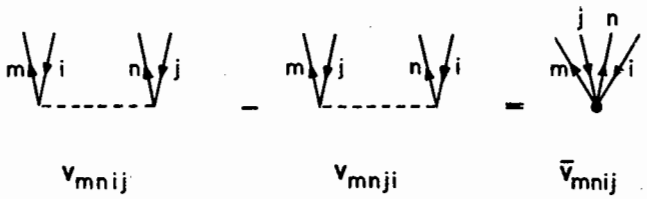


Figure 8.10. Graphical representation of the matrix elements B in Equation (8.71).

As we can see by looking at matrix B of Eq. (8.71), the RPA contains, in addition to the graphs of Fig. 8.3, the graphs shown in Fig. 8.10, which have the meaning of virtual $2p-2h$ excitations.

In general, we can say that the quasi-boson approximation (8.68) is valid for very collective states, that is, if we have many coefficients X^{ν}_{mi} of the same order of magnitude. In such cases each single *ph*-component has only a small probability of being excited and the violation of the Pauli principle can be neglected. We will discuss this point again later in Chapter 9.

On the other hand, the amplitudes Y^{ν}_{mi} should be small compared to the coefficients X^{ν}_{mi} because they describe ground state correlations. If they are too strong, the replacement of the correlated ground state $|\mathrm{RPA}\rangle$ by the $|\mathrm{HF}\rangle$ state in Eq. (8.67) is not justified.

In cases where the coefficients Y^{ν}_{mi} can be neglected completely, we get back to the TDA. This is the reason that we can describe collective states very well using the RPA, where the quasi-boson approach is justified, and also rather pure *ph*-states, where the coefficients Y^{ν}_{mi} are negligible. The RPA equation (8.70) looks like the diagonalization of the Hamiltonian in a basis with a metric tensor $\begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$ which is no longer positive definite. In fact, in Section 8.4.4 we will show that the eigensolutions of Eq. (8.70) are orthogonal with respect to this metric. Another consequence of this metric is that the eigenvalues $\hbar\Omega_{\nu}$ are not necessarily real.

To calculate transition probabilities (B.62) between the excited state $|\nu\rangle$ and the ground state $|0\rangle$ we only need matrix elements of the type $\langle 0|F|\nu\rangle$ for a Hermitian one-body operator F.

In the RPA approximation they are given by

$$\langle 0|F|\nu\rangle = \sum_{kk'} F_{kk'}\rho^{(1)}_{k'k} = \sum_{mi} F_{im}X^{\nu}_{mi} + F_{mi}Y^{\nu}_{mi}.$$

In the following sections we will frequently use the notation

$$\langle 0|F|\nu\rangle = f^{+}\mathcal{X}^{\nu} \tag{8.72}$$

with the column vectors

$$f = \begin{pmatrix} F_{mi} \\ F^{*}_{mi} \end{pmatrix} \qquad \text{and} \qquad \mathcal{X}^{\nu} = \begin{pmatrix} X^{\nu}_{mi} \\ Y^{\nu}_{mi} \end{pmatrix}.$$

8.4.2 Stability of the RPA

The matrix

$$\mathcal{S} = \begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix} \tag{8.73}$$

is exactly the *stability matrix* of the HF theory derived in Eq. (7.37). It is a Hermitian matrix and its real eigenvalues characterize the nature of the HF solution. If $\mathcal{S}$ is positive definite it corresponds to a minimum in the energy surface. In this case, we can calculate the square root of the matrix $\mathcal{S}$ and bring the eigenvalue problem (8.70) into a more symmetric form:

$$\mathcal{S}^{1/2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\mathcal{S}^{1/2}\begin{pmatrix} \tilde{X} \\ \tilde{Y} \end{pmatrix} = \begin{pmatrix} \tilde{X} \\ \tilde{Y} \end{pmatrix}\hbar\Omega$$

with

$$\begin{pmatrix} \tilde{X} \\ \tilde{Y} \end{pmatrix} = \mathcal{S}^{1/2}\begin{pmatrix} X \\ Y \end{pmatrix},$$

which is a Hermitian eigenvalue problem having only real solutions. In other words, if the HF solution corresponds to a minimum in the energy surface and not to a saddle point or a maximum, then the corresponding RPA equation has only real frequencies [Th 61a]. The opposite is not necessarily true. In the following we will assume that we only have real non-vanishing eigenvalues Ω_ν.

8.4.3 Normalization and Closure Relations

Since the RPA matrix is not Hermitian, its eigenvectors cannot be orthogonal in the usual sense. We would also expect a different kind of orthogonality relations from the condition, that the excited states $|\nu\rangle = Q_\nu^+|\mathrm{RPA}\rangle$ should be mutually orthogonal

$$\langle\nu|\nu'\rangle = \delta_{\nu\nu'} = \langle\mathrm{RPA}|[\,Q_\nu, Q_{\nu'}^+\,]|\mathrm{RPA}\rangle \simeq \langle\mathrm{HF}|[\,Q_\nu, Q_{\nu'}^+\,]|\mathrm{HF}\rangle$$

or

$$\delta_{\nu\nu'} = \sum_{mi}\left(X_{mi}^{\nu^*}X_{mi}^{\nu'} - Y_{mi}^{\nu^*}Y_{mi}^{\nu'}\right). \tag{8.74}$$

In fact, we can show that the solutions of the RPA equations for $\nu \neq \nu'$ fulfill this relation exactly. For $\nu = \nu'$ this gives us the possibility of normalizing the vector (X^ν, Y^ν) if the norm is positive [Th 61a].

To show the orthogonality of the RPA solutions in the sense of (8.74) we first notice that to each eigenvector (X^ν, Y^ν) with eigenvalue Ω_ν we have another eigenvector (Y^{ν^*}, X^{ν^*}) with the eigenvalue $-\Omega_\nu$. Both eigenvectors have the same absolute norm in the sense of (8.74) but with different sign.

It is useful to introduce the matrices

$$\mathcal{X}=\begin{pmatrix} X & Y^* \\ Y & X^* \end{pmatrix}; \qquad \mathcal{R}=\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{8.75}$$

Together with (8.75), the RPA equation (8.70) has the form

$$\mathcal{S}\mathcal{X}=\mathcal{R}\mathcal{X}\Omega, \tag{8.76}$$

where the diagonal matrix Ω contains the real eigenvalues $(\hbar\Omega_\mu, -\hbar\Omega_\mu)$. Simple matrix algebra shows that

$$[\Omega, \mathcal{X}^+\mathcal{R}\mathcal{X}]=(\mathcal{R}\mathcal{X}\Omega)^+\mathcal{X}-\mathcal{X}^+(\mathcal{R}\mathcal{X}\Omega)=\mathcal{X}^+(\mathcal{S}^+-\mathcal{S})\mathcal{X}=0 \tag{8.77}$$

that is, Ω commutes with $\mathcal{X}^+\mathcal{R}\mathcal{X}$, and thus $\mathcal{X}^+\mathcal{R}\mathcal{X}$ is diagonal together with Ω. Since the norm of the vectors (X, Y) is open, we choose*

$$\mathcal{X}^+\mathcal{R}\mathcal{X}=\mathcal{R}. \tag{8.78}$$

These are exactly the orthogonality conditions (8.74).

The closure condition is obtained by multiplying (8.78) with $\mathcal{R}$, which shows that $\mathcal{R}\mathcal{X}\mathcal{R}$ is the inverse matrix of $\mathcal{X}^+$, or

$$\mathcal{X}\mathcal{R}\mathcal{X}^+=\mathcal{R} \tag{8.79}$$

which gives explicitly

$$\sum_\nu X^\nu_{mi}X^{\nu*}_{m'i'}-Y^{\nu*}_{mi}Y^\nu_{m'i'}=\delta_{mm'}\delta_{ii'}. \tag{8.80}$$

8.4.4 Numerical Solution of the RPA Equations

In many practical cases the RPA-matrices A and B are real. For real frequences Ω_ν the problem can be reduced to the diagonalization of a real *symmetric* matrix of *half the dimension*.

We define the vectors (see footnote page 301)

$$P^\nu=i\cdot\sqrt{\frac{\hbar\Omega_\nu}{2}}\,(X^\nu+Y^\nu); \qquad Q^\nu=\sqrt{\frac{\hbar}{2\Omega_\nu}}\,(X^\nu-Y^\nu). \tag{8.81}$$

From (8.70) it is easy to deduce

$$\begin{aligned} i\cdot(A-B)Q^\nu&=\hbar P^\nu,\\ -i\cdot(A+B)P^\nu&=\hbar\Omega_\nu^2 Q^\nu, \end{aligned} \tag{8.82}$$

and

$$(A+B)(A-B)Q^\nu=\hbar^2\Omega_\nu^2 Q^\nu. \tag{8.83}$$

Together with the stability matrix $\mathcal{S}$ (8.73), the matrices $(A\pm B)$ are positive definite. We can decompose $A-B$ into a product of two triangular matrices (Orthogonalization of Gram–Schmidt [Wi 65, Ch 70], we

* For a positive definite stability matrix $\mathcal{S}$ this is always possible, because Ω_ν then has the same sign as $(\mathcal{X}^+\mathcal{R}\mathcal{X})_{\nu\nu}$.

could also search for the square root of the matrix $(A-B)$ [UR 71])

$$(A-B)=T^{T}T \qquad \text{with } T_{ik}=0 \text{ for } i>k \tag{8.84}$$

and be left with the real symmetric eigenvalue problem

$$T(A+B)T^{T}R^{\nu}=\hbar^{2}\Omega_{\nu}^{2}R^{\nu}. \tag{8.85}$$

Its solution gives the eigenvalues Ω_{ν} and the normalized eigenvectors R^{ν}. Finally, we get the properly normalized RPA amplitudes from

$$\begin{pmatrix} X^{\nu} \\ Y^{\nu} \end{pmatrix}=\frac{1}{2}\left((\hbar\Omega_{\nu})^{-1/2}T^{T}R^{\nu}\pm(\hbar\Omega_{\nu})^{1/2}T^{-1}R^{\nu}\right). \tag{8.86}$$

8.4.5 Representation by Boson Operators

For the derivation of the RPA equations (8.70) we have used the quasi-boson approximation (8.68), that is, we have replaced the *ph* operators $a_m^{+}a_i$ by boson operators B_{mi}^{+}

$$a_m^{+}a_i \rightarrow B_{mi}^{+}, \qquad a_i^{+}a_m \rightarrow B_{mi}, \tag{8.87}$$

which fulfill exact boson commutation relations

$$[B_{mi}, B_{m'i'}]=[B_{mi}^{+}, B_{m'i'}^{+}]=0; \qquad [B_{mi}, B_{m'i'}^{+}]=\delta_{ii'}\delta_{mm'}. \tag{8.88}$$

This replacement is an approximation. In fact we will see in Chapter 9 that we can expand the Fermion pair operators $a_m^{+}a_i$ in a series of boson operators, in which the first term is B_{mi}^{+}. In the quasi-boson approximation we neglect all higher terms.

Since we use this approximation within the RPA theory it is very convenient to represent other operators, such as the Hamiltonian H or transition operators, by the B^{+}s and Bs. How this can be done is discussed in Chapter 9.

Here we shall go to harmonic order only, that is, we take into account in the Hamiltonian H_B only terms quadratic in B, B^{+} as $B^{+}B$, BB, and $B^{+}B^{+}$. This is consistent with the approach (8.87) because we need only terms of the form $[H, B^{+}], [H, B]$ (see Eq. 8.67) in the derivation of the RPA equations. In particular, we determine the coefficients of the terms $B^{+}B$, BB, and $B^{+}B^{+}$ of the Hamiltonian by the requirement* (8.71):

$$\langle \mathrm{HF}|\left[B_{mi}, \left[H_B, B_{nj}^{+}\right]\right]|\mathrm{HF}\rangle=\langle \mathrm{HF}|\left[a_i^{+}a_m, \left[H, a_n^{+}a_j\right]\right]|\mathrm{HF}\rangle=A_{minj},$$

$$\langle \mathrm{HF}|\left[B_{mi}, \left[H_B, B_{nj}\right]\right]|\mathrm{HF}\rangle=\langle \mathrm{HF}|\left[a_i^{+}a_m, \left[H, a_j^{+}a_n\right]\right]|\mathrm{HF}\rangle=-B_{minj}, \tag{8.89}$$

*To be precise, we must remark that $|\mathrm{HF}\rangle$ is not exactly the ground state of the Boson operators B_{mi}^{+}, since they coincide only approximately with the operators $a_i^{+}a_m$. In Chapter 9, we define the ground states of Boson operators by round parenthesis $B_{mi}|\mathrm{HF})=0$. There is no difference between $|\mathrm{HF}\rangle$ and $|\mathrm{HF})$ in RPA order; we do not distinguish between these two states in the present chapter and always use $B_{mi}|\mathrm{HF}\rangle=0$.

and find*

$$H_B = E_{\mathrm{HF}} + \sum_{minj} A_{minj} B^+_{mi} B_{nj} + \frac{1}{2} \sum_{minj} (B_{minj} B^+_{mi} B^+_{nj} + \mathrm{h.c.})$$
$$= E_{\mathrm{HF}} - \frac{1}{2} \sum_{mi} A_{mimi} + \frac{1}{2} (B^+ \quad B) \begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix} \begin{pmatrix} B \\ B^+ \end{pmatrix}. \tag{8.90}$$

The constant E_{HF} corresponds to the expectation value of the Hamiltonian in the HF ground state, as can be seen by switching Eq. (8.90) between $|\mathrm{HF}\rangle$. In Chapter 9, it is shown that the higher order terms in B give no contributions. There are no linear terms in the operators B, B^+ because we start with a self-consistent solution [$H^{20} = 0$, Eq. (7.36)]. A Hamiltonian of this form can also be derived in second order by the method of generator coordinates, which has been introduced in this context by Jancovici and Schiff [JS 64] (see also Chap. 10).

The diagonalization of (8.90) is straightforward if we introduce a Bogoliubov transformation [see (E.60)] amongst the Boson operators corresponding to (8.65) and replace the operators Q_ν by pure Boson operators O_ν.

$$O^+_\nu = \sum_{mi} X^\nu_{mi} B^+_{mi} - Y^\nu_{mi} B_{mi}. \tag{8.91}$$

The requirement that the operators O^+_ν, O_ν now fulfill exact boson commutation relations is equivalent to the unitarity of this transformation and immediately yields the orthogonality relations (8.78) and (8.79).

The Hamiltonian expressed in terms of the operators O^+_ν, O_ν is found to be diagonal after taking into account the RPA equation (8.76) viz:

$$H_B = E_{\mathrm{HF}} - \frac{1}{2} \mathrm{Tr}\, A + \frac{1}{2} (O^+ \quad O) \mathcal{X}^+ \mathcal{S} \mathcal{X} \begin{pmatrix} O \\ O^+ \end{pmatrix}$$
$$= E_{\mathrm{RPA}} + \sum_\nu \hbar \Omega_\nu O^+_\nu O_\nu \tag{8.92}$$

with

$$E_{\mathrm{RPA}} = E_{\mathrm{HF}} - \frac{1}{2} \mathrm{Tr}\, A + \frac{\hbar}{2} \sum_{\nu > 0} \Omega_\nu \tag{8.93a}$$
$$= E_{\mathrm{HF}} - \sum_\nu \hbar \Omega_\nu \sum_{mi} |Y^\nu_{mi}|^2. \tag{8.93b}$$

In Eq. (8.93) we have used the inverse of the RPA equation (8.76)

$$A = X \Omega X^+ + Y^* \Omega Y^T,$$
$$-B = X \Omega Y^+ + Y^* \Omega X^T. \tag{8.94}$$

The Hamiltonian (8.92) corresponds to the Hamiltonian of harmonic oscillators. Therefore, the RPA is called the harmonic approximation. It

*We should not, in the following matrix notation, confuse the operators B_{mi} with the matrix B_{minj}.

determines uncoupled eigenmodes of the system. The eigenfunctions of H_B are the ground state $|\mathrm{RPA}\rangle$ (it will be constructed in Section 8.4.6), one-boson states $O_\nu^+|\mathrm{RPA}\rangle$, two-boson states $O_\nu^+ O_{\nu'}^+|\mathrm{RPA}\rangle$, and so on.

Since A is just the matrix, which is diagonalized in the TDA equation (8.10), we see from Eq. (8.93) that the RPA ground state energy E_{RPA} is lowered compared to the Hartree–Fock energy E_{HF} by the amount

$$E_{\mathrm{HF}} - E_{\mathrm{RPA}} = \frac{1}{2} \sum_{\nu>0} (E_\nu^{\mathrm{TDA}} - E_\nu^{\mathrm{RPA}}).$$

It is just half the sum of all differences in the excitation energies between the RPA and the TDA approach.

As we know from the theory of the harmonic oscillator, we may also represent the eigenmodes by generalized coordinates $\mathcal{P}_\nu$ and moments $\mathcal{Q}_\nu$ defined by [MW 69b]

$$\mathcal{P}_\nu = \frac{\hbar}{i} \sqrt{\frac{M_\nu \Omega_\nu}{\hbar}} \frac{1}{\sqrt{2}} (O_\nu - O_\nu^+); \qquad \mathcal{Q}_\nu = \sqrt{\frac{\hbar}{M_\nu \Omega_\nu}} \frac{1}{\sqrt{2}} (O_\nu + O_\nu^+). \tag{8.95}$$

The numbers M_ν are arbitrary at this point. The operators $\mathcal{P}_\nu, \mathcal{Q}_\nu$ fulfill the commutation relations of conjugate momenta and coordinates

$$[\mathcal{P}_\nu, \mathcal{P}_{\nu'}] = [\mathcal{Q}_\nu, \mathcal{Q}_{\nu'}] = 0; \qquad [\mathcal{P}_\nu, \mathcal{Q}_{\nu'}] = \frac{\hbar}{i} \delta_{\nu\nu'}. \tag{8.96}$$

The Hamiltonian (8.92) can be expressed in these operators:

$$H_B = E_{\mathrm{HF}} - \frac{1}{2} \mathrm{Tr}\, A + \sum_\nu \left(\frac{1}{2M_\nu} \mathcal{P}_\nu^2 + \frac{M_\nu}{2} \Omega_\nu^2 \mathcal{Q}_\nu^2 \right). \tag{8.97}$$

The operators $\mathcal{P}_\nu, \mathcal{Q}_\nu$, therefore, obey the equations of motion

$$[H_B, \mathcal{P}_\nu] = i\hbar \Omega_\nu^2 M_\nu \mathcal{Q}_\nu; \qquad [H_B, \mathcal{Q}_\nu] = -\frac{i\hbar}{M_\nu} \mathcal{P}_\nu. \tag{8.98}$$

Using the representation

$$\mathcal{P}_\nu = \sum_{mi} P_{mi}^\nu B_{mi}^+ + P_{mi}^{\nu *} B_{mi} \qquad \text{with } P_{mi}^\nu = i\hbar \sqrt{\frac{M_\nu \Omega_\nu}{2\hbar}} \, (X + Y^*)_{mi}^\nu ,$$

$$\mathcal{Q}_\nu = \sum_{mi} Q_{mi}^\nu B_{mi}^+ + Q_{mi}^{\nu *} B_{mi} \qquad \text{with } Q_{mi}^\nu = \sqrt{\frac{\hbar}{2M_\nu \Omega_\nu}} \, (X - Y^*)_{mi}^\nu ,$$

we get an explicit form for the Eqs. (8.98):

$$\begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix} \begin{pmatrix} P \\ -P^* \end{pmatrix}_\nu = i\hbar \Omega_\nu^2 M_\nu \begin{pmatrix} Q \\ Q^* \end{pmatrix}_\nu ;$$
$$\begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix} \begin{pmatrix} Q \\ -Q^* \end{pmatrix}_\nu = \frac{\hbar}{i} \frac{1}{M_\nu} \begin{pmatrix} P \\ P^* \end{pmatrix}_\nu \tag{8.99}$$

Neither the normalization of the vectors P^{ν}_{mi} and Q^{ν}_{mi} nor the parameter M_ν is determined from the solution of this equation. The condition (8.96),

$$(P^* \quad P)_\nu \begin{pmatrix} Q \\ -Q^* \end{pmatrix}_{\nu'} = \frac{\hbar}{i}\delta_{\nu\nu'}, \tag{8.100}$$

can be used to determine either the parameter M_ν (for fixed normalization of P^{ν}_{mi} or Q^{ν}_{mi}) or the normalization of these vectors (for fixed parameter M_ν).

So far we have only treated the case of non-vanishing real RPA frequencies ($\Omega^2_\nu > 0$). In principle, the RPA matrix can also have complex eigenvalues. (An example is $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, with the eigenvalues $\Omega_\nu = \pm 1 \pm i$.) Until now such general cases have not been investigated. In the context of adiabatic time-dependent Hartree–Fock theory (see Sec. 12.3), it happens sometimes that we must diagonalize a matrix of the RPA type with a pure imaginary eigenvalue $\Omega_i = i|\Omega_i|$. The corresponding eigenvector $\mathcal{X}_i$ has zero norm:

$$i|\Omega_i|\mathcal{X}_i^+ \mathcal{N}\mathcal{X}_i = \mathcal{X}_i^+ \mathcal{S}\mathcal{X}_i = (\mathcal{X}_i^+ \mathcal{S}\mathcal{X}_i)^+ = -i|\Omega_i|\mathcal{X}_i^+ \mathcal{N}\mathcal{X}_i.$$

In such a case there exists no boson of the form (8.91), which corresponds to this eigenvalue. Nevertheless, we can construct operators $\mathcal{P}_\nu$ and $\mathcal{Q}_\nu$ with the properties (8.96) and (8.98). We have only to solve the system (8.99) for an arbitrary positive value of M_ν and a negative value of Ω_i^2. In order to obtain a complete set of boson operators, we can then define corresponding bosons O_i, O_i^+ by Eq. (8.95) using the absolute value $|\Omega_i|$ instead of Ω_i [RB 76]. The Hamiltonian H_B (8.97) is not diagonal in this basis, but "maximally" off diagonal:

$$-\frac{\hbar}{2}|\Omega_i|(O_i^+ O_i^+ + O_i O_i).$$

8.4.6 Construction of the RPA Ground State

The RPA ground state was originally defined by Eq. (8.66) as the vacuum of the operators Q_ν. Within the quasi-boson approximation it is now equivalent to the vacuum $|\mathrm{RPA}\rangle$ of the boson operators O_ν;

$$O_\nu|\mathrm{RPA}\rangle = 0.$$

We can construct it explicitly from the vacuum $|\mathrm{HF}\rangle$ of the bosons B_{mi} [Eq. (8.87)] in analogy to Chapter 7, where we constructed the vacuum of quasi-particle operators, by the theorem of Thouless [Th 60] (see also Sec. E.5):

$$|\mathrm{RPA}\rangle = N_0 e^{\hat{Z}}|\mathrm{HF}\rangle \tag{8.101}$$

where

$$\hat{Z} = \frac{1}{2}\sum_{minj} Z_{minj} B^+_{mi} B^+_{nj}$$

and N_0 is a normalization (see Eqs. E.69 and E.75). In Appendix E we get, for the matrix Z, from (8.91) and the commutation relations (8.88),

$$Z = Y^* X^{*-1}.$$

The inversion of the matrix X is, in principle, possible, but more practicable methods have been proposed by Sanderson [Sa 65] and by da Providencia [Pr 66]. Actual calculations have been performed by Goswanni and Pal [GP 63]. It turns out that the RPA ground state has to be taken with care, since the quasi-boson approximation probably overestimates the ground state correlations in many cases [IUY 65, PR 68, UR 69].

The energy of the ground state is given by Eq. (8.93). Obviously, it is always lower than the Hartree–Fock energy E_{HF}. The reason is that it takes into account higher correlations. However, we have to keep in mind that the RPA does not follow from a variational principle. It may, therefore, happen that its energy can even be deeper than the exact energy.

Since we are now able to calculate the RPA ground state, we could use it to evaluate Eq. (8.67), avoiding the quasi-boson approximation (8.68) in the calculation of the matrices A and B. Diagonalizing the new RPA equations gives a better ground state, and so on. This yields a self-consistent prescription which avoids the shortcomings of the quasi-boson approximation [Ro 70, SE 73, Ma 76a]. We will come to this method in more detail in Section 9.2.3.

Over the years, several other extensions of the RPA theory have been proposed (so-called *higher random phase approximations*; see, for instance, [Sa 62, TU 64, Ro 68a, MYM 68, GNS 70, DDK 71]). We do not want to discuss these methods in the framework of this book, since in Chapter 9 we present a systematic way of treating collective phenomena in nuclei, the so-called boson expansion technique.

8.4.7 Invariances and Spurious Solutions

We assume in the following that the exact two-body Hamiltonian H is invariant under a continuous symmetry operation generated by a Hermitian one-body operator $\hat{P}$, as there is, for instance, the case of translation; we assume further that the HF solution violates this symmetry. This means, in particular, that the HF single-particle density $\rho^{(0)}$ does not commute with $\hat{P}$

$$\left[\rho^{(0)}, \hat{P}\right] \neq 0. \tag{8.102}$$

Since $\rho^{(0)}$ is diagonal in the HF basis (0 for particles and 1 for holes) this means that not all the *ph* matrix elements P_{mi} vanish.

The exact Hamiltonian commutes with $\hat{P}$:

$$\left[H, \hat{P}\right] = 0. \tag{8.103}$$

$\hat{P}$ is therefore an exact but *spurious solution* of the RPA equation $\langle \mathrm{HF} | [\delta O, [H, \hat{P}]] | \mathrm{HF} \rangle = 0$* or, in matrix language,

$$\begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix} \begin{pmatrix} P \\ -P^* \end{pmatrix} = 0, \tag{8.104}$$

* The *pp* and *hh* matrix elements of $\hat{P}$ do not contribute to this equation because of (5.35)

where P is the vector P_{mi} in particle-hole space. The corresponding state is

$$|P\rangle = \sum_{mi} (P_{mi} a_m^+ a_i + P_{mi}^* a_i^+ a_m)|\mathrm{RPA}\rangle. \tag{8.105}$$

We realize that the RPA equation (8.70) has such a solution only for the case in which the symmetry is, in fact, broken by the HF solution. Otherwise the matrix elements P_{mi} vanish identically.

Thus, we see that to the extent that we calculate the RPA exactly and use self-consistent single-particle energies and wave functions* the spurious excitations that correspond to a broken symmetry in the HF state—as, for example, in translation of the nucleus as a whole—separate out. They are orthogonal to the other excitations and lie at zero excitation energy. We have derived this for density-independent interactions. However, in Section (8.5) we will see that this also holds for density-dependent interactions that obey the condition (5.71).

This fact is a major advantage of the RPA over the TDA where, as we have seen in Section 8.3.3, the spurious and physical solutions become mixed up with one another.

As we stated at the beginning of Section 8.4.3, the RPA always has two symmetric solutions, Ω_ν and $-\Omega_\nu$. For $\hat{P}$, however, we have no adjoint partner, since $\hat{P}$ is Hermitian and the adjoint is simply a repetition of itself. In fact, the state $|P\rangle$ in Eq. (8.105) is not normalizible in the sense of Eq. (8.74). There exists no corresponding boson. Since the RPA matrix has even dimensions and all non-spurious solutions are paired off, the RPA solutions are one short of forming a complete set.

To see what is going on, we follow Marshalek and Weneser [MW 69b, MW 70], and use the representation (8.95) for the spurious state. Obviously, the *ph* and the *hp* parts of $\hat{P}$ are now identical to the Hermitian operator $\mathcal{P}_0$ (the canonical momentum) up to a constant M_0 which, in the definition (8.95) is open in any case. In order to obtain complete sets of operators O_ν^+, O_ν $(\nu > 0)$ and $\mathcal{P}_0, \mathcal{Q}_0$ we only have to determine the canonical coordinate $\mathcal{Q}_0$. It is given by the solution of Eq. (8.99):

$$\begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix} \begin{pmatrix} Q \\ -Q^* \end{pmatrix}_0 = -\frac{i\hbar}{M_0} \begin{pmatrix} P \\ P^* \end{pmatrix}_0, \tag{8.106}$$

which is a linear inhomogeneous problem. Since the matrix $\mathcal{S}$ is singular, this equation only has a solution if the inhomogeneous part is perpendicular to the spurious solution. But this is exactly the case (8.104). The equations (8.106) were first derived by Thouless and Valatin in the context of nuclear rotations [TV 62].

The constant M_0 is finally determined by the commutation relation (8.96):

$$\sum_{mi} (P_{mi}^* Q_{mi} - P_{mi} Q_{mi}^*)_0 = (P^* \quad P)_0 \begin{pmatrix} Q \\ -Q^* \end{pmatrix}_0 = \frac{\hbar}{i}. \tag{8.107}$$

* In a case where this is not true, for instance, in RPA calculations based on a phenomenological Woods–Saxon potential, special methods have been developed to remove the spurious components [PS 77, Me 79].

Inverting (8.106), we find

$$\frac{\hbar^2}{2M_0}=\frac{1}{2}(Q^* \quad -Q)_0\begin{pmatrix}A & B\\ B^* & A^*\end{pmatrix}\begin{pmatrix}Q\\ -Q^*\end{pmatrix}_0=\frac{1}{2}\langle \mathrm{HF}|[\mathcal{Q}_0,[H,\mathcal{Q}_0]]|\mathrm{HF}\rangle \tag{8.108}$$

In practical cases it is often possible to choose the phases in such a way that A and B are real. In a case where Q_{mi} is also real, P_{mi} has to be purely imaginary and we get, for instance,

$$M_0=2P_0^*(A-B)^{-1}P_0. \tag{8.109}$$

We can now write down the Hamiltonian H_B in RPA order in the following way (8.97).

$$H_B=E_{\mathrm{RPA}}+\sum_{\nu>0}\hbar\Omega_\nu O_\nu^+ O_\nu+\frac{\mathcal{P}_0^2}{2M_0}. \tag{8.110}$$

The constant E_{RPA} no longer has the form (8.93b) because of the zero frequency mode. However, we can give it easily by constructing the HF expectation value of (8.110):

$$E_{\mathrm{RPA}}=E_{\mathrm{HF}}-\sum_{\nu>0}\hbar\Omega_\nu\sum_{mi}|Y_{mi}^\nu|^2-\frac{\langle \mathrm{HF}|\mathcal{P}_0^2|\mathrm{HF}\rangle}{2M_0}. \tag{8.111}$$

The subtraction of the term $\langle\hat{P}^2\rangle/2M_0$ is often used in the recipes (Chap. 11) for doing HF calculations in nuclei in order to correct for spurious translational motion of the centre of mass. We see that this correction is included automatically in the RPA.

The form of the operator (8.110) also gives us an interpretation of the constant M_0. It is the inertial parameter corresponding to the motion characterized by $\hat{P}$.

Let us consider two examples:

(i) *Translation.* Here we can use the Galilean invariance for the exact Hamiltonian

$$[H,\mathbf{X}]=\frac{-i\hbar}{Am}\mathbf{P}, \tag{8.112}$$

where $\mathbf{X}$ is the center-of-mass coordinate, and find by calculating the matrix elements $\langle \mathrm{HF}|[a_i^+ a_m\ldots]|\mathrm{HF}\rangle$ of this equation an equation identical to (8.106). This shows that in this case $M_0=Am$ is the total mass of the nucleus.

(ii) *Rotation.* Starting with a deformed HF calculation, a violated symmetry is a rotation perpendicular to the symmetry axis, let us say a rotation around the x-axis—the operator $\hat{P}$ is now the angular momentum operator $\hat{J}_x$. Equation (8.106) determines the *ph*-matrix element of a corresponding coordinate, an angle $\hat{\theta}_x$. M_0 is now a moment of inertia $\mathcal{I}_{\mathrm{TV}}$ corresponding to a rotation around the x-axis. We get for real matrix elements $J_{x_{mi}}$ in analogy to Eq. (8.109)

$$\mathcal{I}_{\mathrm{TV}}=2\sum_{mi}J_{x_{mi}}^*(A+B)^{-1}_{minj}J_{x_{nj}}. \tag{8.113}$$

with

$$(A+B)_{minj}=\delta_{mn}\delta_{ij}(\epsilon_m-\epsilon_i)+\bar{v}_{mjin}+\bar{v}_{mnij}.$$

Neglecting the residual interaction $\bar{v}$ we again find the well-known cranking formula (3.89). For a discussion of the numerical application of this formula see Chap. (3.4.2).

In the next step, we would like to construct wave functions corresponding to these "spurious" excitations. We can define a ground state $|0\rangle$ by

$$O_\nu|0\rangle=0 \qquad \text{for } \nu>0; \qquad \mathcal{P}_0|0\rangle=0;$$

and excited states

$$|\nu\rangle=O_\nu^+|0\rangle \qquad \text{for } \nu>0; \qquad |p\rangle=\exp\left(i\frac{p}{\hbar}\mathcal{Q}_0\right)|0\rangle.$$

These wave functions give the proper excitation energies Ω_ν and $p^2/2M_0$, but it turns out that they are no longer normalizible. [MW 70]. The reason for this is that the RPA is a small amplitude approximation. Therefore, we describe the wave function of a rotational state, for instance, properly only for small angles. We can deduce therefore, energies, but not the full wave function. In particular, we do not recover the quantization of the angular momentum.

Summarizing, we can say that the RPA theory treats the inherent symmetries of the problem consistently. It separates the so-called spurious motions from the vibrations. In fact, these excitations are not really spurious, but they represent a different type of motion which has to be treated separately. At least their energies (which are not given by $\Omega_0=0$ but by $p^2/2M_0$) are reproduced properly, since they are characterized by the right mass parameters.

8.5 Linear Response Theory

In calculating collective excitations of the nuclear system, we have so far used the stationary Schrödinger equation (8.59) and tried to diagonalize the Hamiltonian at least in some approximation.

We shall now begin from quite a different starting point. We investigate the influence of an external time-dependent field

$$F(t)=Fe^{-i\omega t}+F^+e^{i\omega t} \tag{8.114}$$

on the system. We assume that F is a one-body operator, that is,

$$F(t)=\sum_{kl} f_{kl}(t)a_k^+a_l,$$

and that the field is weak, that is, it introduces only small changes of the nuclear density, which we can treat in linear order.

As we will see, the nuclear density oscillates with this external field and we obtain resonances whenever the frequency ω is close to an excitation energy of the system. In this way:

(i) We get information about the excited states of the system. In particular, we find an equation for the amplitudes $\langle 0|a_i^+ a_m|\nu\rangle$, which corresponds precisely to the RPA equations.
(ii) We are able to calculate the response of the system to the external field, that is, changes in its density, energy, and so on.
(iii) We find a method to derive the RPA equations for density-dependent forces.

8.5.1 Derivation of the Linear Response Equations

The wave function $|\Phi(t)\rangle$ of a nuclear system in an external, time-dependent field is no longer stationary. It is a wave packet, and its one-body density

$$\rho_{kl}(t)=\langle\Phi(t)|a_l^+ a_k|\Phi(t)\rangle \tag{8.115}$$

is now time dependent. We want to calculate this density explicitly under the following approximations.

(i) We assume that at any time $\rho(t)$ corresponds to a *Slater determinant** (i.e., $\rho^2=\rho$). Then ρ obeys the following equation of motion.

$$i\hbar\dot{\rho}=\big[h[\rho]+f(t),\rho\big]. \tag{8.116}$$

This is the *time-dependent Hartree–Fock (TDHF) equation*. It will be derived in Section 12.2. Here $h[\rho]$ is the single-particle Hartree–Fock (HF) field as defined in Eq. (5.33) and f is the time-dependent external field of Eq. (8.114).

(ii) We assume that the external field $f(t)$ is weak, that is, it introduces only oscillations with *small amplitudes* around the stationary density $\rho^{(0)}$, which is itself a solution of the stationary Hartree–Fock equation (5.36) $[h[\rho^{(0)}],\rho^{(0)}]=0$. Therefore, the density has the form

$$\rho(t)=\rho^{(0)}+\delta\rho(t), \tag{8.117}$$

where

$$\delta\rho=\rho^{(1)}e^{-i\omega t}+\rho^{(1)^+}e^{i\omega t} \tag{8.118}$$

is linear in the field f. In the following we work in the basis in which $\rho^{(0)}$ and $h[\rho^{(0)}]$ are diagonal, that is, in the HF-basis:

$$\rho_{kl}^{(0)}=\delta_{kl}\cdot\rho_k^{(0)}=\begin{cases}0 & \text{for particles,}\\ 1 & \text{for holes,}\end{cases} \tag{8.119}$$

and

$$(h_0)_{kl}=\big(h\big[\rho^{(0)}\big]\big)_{kl}=\delta_{kl}\cdot\epsilon_k. \tag{8.120}$$

* In this sense, the following derivation of the RPA shows that it is just a time dependent generalization of the independent particle picture (see Chap. 12).

In Section 5.3.3 we saw that the condition $\rho^2=\rho$ implies that the only non-vanishing matrix elements of $\rho^{(1)}$ are *ph* and *hp* matrix elements $\rho^{(1)}_{mi}$ and $\rho^{(1)}_{im}$. They are determined by the solution of the TDHF equation (8.116). We insert Eq. (8.117) and expand up to linear order in the external field f,

$$i\hbar\delta\dot{\rho}=[h_0,\delta\rho]+\left[\frac{\delta h}{\delta\rho}\cdot\delta\rho,\rho^{(0)}\right]+[f,\rho^{(0)}], \tag{8.121}$$

where $\delta h/\delta\rho\cdot\delta\rho$ is a shorthand notation for

$$\sum_{im}\left(\left.\frac{\partial h}{\partial\rho_{mi}}\right|_{\rho=\rho^{(0)}}\cdot\delta\rho_{mi}+\left.\frac{\partial h}{\partial\rho_{im}}\right|_{\rho=\rho^{(0)}}\cdot\delta\rho_{im}\right). \tag{8.122}$$

Using the rules for the calculation with HF densities, as given in Appendix D, [Eq. (D.30)ff] we find that the *pp* and the *hh* matrix elements of Eq. (8.121) vanish identically. From (8.118) we obtain for the *ph* and *hp* elements the *linear response equation*

$$\left\{\begin{pmatrix}A & B\\ B^* & A^*\end{pmatrix}-\hbar\omega\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\right\}\begin{pmatrix}\rho^{(1)ph}\\ \rho^{(1)hp}\end{pmatrix}=-\begin{pmatrix}f^{ph}\\ f^{hp}\end{pmatrix} \tag{8.123}$$

with

$$A_{minj}=(\epsilon_m-\epsilon_i)\delta_{mn}\delta_{ij}+\frac{\partial h_{mi}}{\partial\rho_{nj}}\,;\qquad B_{minj}=\frac{\partial h_{mi}}{\partial\rho_{jn}}\,.$$

These matrices correspond exactly to the matrices A and B of the RPA method, if we use as a residual interaction [see Eq. (5.32)]

$$\tilde{v}_{psqr}=\frac{\partial h_{pq}}{\partial\rho_{rs}}=\frac{\partial^2 E}{\partial\rho_{qp}\partial\rho_{rs}}\,. \tag{8.124}$$

In the case of HF theory without density dependent forces, we can use the expression (5.28) for the energy and thus we get back the RPA matrices (8.71). However, the above derivation is more general. It can also be applied to theories with *density dependent forces* (see Sec. 5.6). In this case, for the calculation of excited states we have to use the force defined by Eq. (8.124) as the second derivative of the ground state energy with respect to the density. In particular, this force is no longer necessarily antisymmetric in the indices q and r.

The linear response equation (8.123) is an inhomogeneous equation and can be solved by inverting the matrix on the left-hand side. We then find a linear connection between the external field f and the change in the nuclear density (i.e., the response of the system):

$$\rho^{(1)}_{kl}=\sum_{pq}R_{klpq}(\omega)f_{pq}. \tag{8.125}$$

The function $R_{klpq}(\omega)$ is called the *response function* [see Eqs. (F.51) and (F.68)]. We have calculated it here only in the mean field approach (i.e., in "RPA order"), because we restrict ourselves to product wave functions

with $\rho^2=\rho$. Therefore, the indices *kl* and *pq* run only over *ph* and *hp* indices. This is no longer true in the general case.

The response function R depends on the frequency of the external field. It has poles at the eigenfrequencies of the system, where already an infinitesimal field f is sufficient to excite the corresponding eigenmode.

To find these resonances ($\omega=\Omega_\nu$), we have to look for the solutions of the homogeneous equation (8.123) with vanishing external field. With the notation of Eqs. (8.73) and (8.75) we obtain

$$(\mathbb{S}-\hbar\Omega_\nu\mathcal{R})\rho^{(1)\nu}=0. \tag{8.126}$$

This is exactly the RPA equation (8.70). Its solution gives the transition densities (8.69)

$$\rho^{(1)}_{pq}(\Omega_\nu)=\langle 0|a_q^+ a_p|\nu\rangle \tag{8.127}$$

The derivation of the RPA equations is more general than the one given in Section 8.4.1, because it can also be used in the case of density-dependent forces, which is a crucial point for the validity of the mean field approach in nuclear physics.

On the other hand, the derivation shows us also very clearly the connection of the random phase approximation to the Hartree–Fock theory. We allow the average nuclear potential to oscillate around its stationary value, which corresponds to a minimum in the energy surface of all possible product wave functions (see Sec. 7.3.1). In the limit of small amplitudes we thus get a linear eigenvalue problem for the determination of the normal modes of the system (see also Sec. 12.3.2). The *RPA approximation* is therefore nothing but the *small amplitude limit* of the *time-dependent mean field approach.*

The energy surface in the vicinity of the stationary point ρ_0 can be obtained by expanding the HF energy $E[\rho]$, up to second order in $\delta\rho$* [BG 77]:

$$\begin{aligned} E[\rho] &= E[\rho^{(0)}]+\mathrm{Tr}\left(\frac{\delta E}{\delta\rho}\delta\rho\right)+\frac{1}{2}\mathrm{Tr}\,\mathrm{Tr}\left(\delta\rho+\frac{\delta^2 E}{\delta\rho\delta\rho}\delta\rho\right) \\ &= E[\rho^{(0)}]+\frac{1}{2}\rho^{(1)+}\mathbb{S}\rho^{(1)}. \end{aligned} \tag{8.128}$$

We find that the stability matrix $\mathbb{S}$ has again the form (7.37), but now the matrices A and B (8.123) are also defined for density-dependent forces.

It is easy to show also for such forces that we eventually obtain *spurious solutions* at zero frequency in the case of a broken symmetry $\hat{P}$. We have only to use the condition (5.68), which states that

$$\bar{\rho}=e^{i\alpha P}\rho^{(0)}e^{-i\alpha P},$$

* In deriving Eq. (8.128) we must be aware of the fact that $\delta\rho$ has in second order also *pp*- and *hh* matrix elements. They are, however, not independent variables, because any Slater determinant can be expressed by the *ph* and *hp* matrix elements of $\delta\rho$. We can use the relation $\rho^2=\rho$ to eliminate them. Thus, the term $\mathrm{Tr}(h_0\cdot\delta\rho)$ gives a quadratic contribution, namely, just the *ph* energies $\epsilon_m-\epsilon_i$ in the matrix A.

as well as $\rho^{(0)}$, also fulfills the equation

$$[h(\bar{\rho}), \bar{\rho}] = 0.$$

Infinitesimal changes $\bar{\rho} = \rho^{(0)} + \rho^{(1)}$, with $\rho^{(1)} \propto [P, \rho^{(0)}]$, gives

$$\left[h_0, \rho^{(1)}\right] + \left[\frac{\delta h}{\delta \rho} \cdot \rho^{(1)}, \rho^{(0)}\right] = 0, \tag{8.129}$$

which shows that $\rho^{(1)}$, that is, just the *ph* and *hp* matrix elements of P, correspond to a spurious solution at zero energy analogous to Eq. (8.104) in the case of density-independent forces.

Knowing the eigenmodes of the system, that is, the frequencies Ω_ν and the RPA amplitudes X and Y, makes it possible to solve the linear response equation (8.123). Using (8.76ff), we find

$$\mathcal{S} - \hbar\omega\mathcal{N} = \hbar\mathcal{N}\mathcal{X}(\Omega - \omega)\mathcal{N}\mathcal{X}^+\mathcal{N}$$

which can easily be inverted:

$$\rho^{(1)} = \frac{1}{\hbar}\mathcal{X}(\omega - \Omega)^{-1}\mathcal{N}\mathcal{X}^+ f.$$

This is equivalent to Eq. (8.125)—we have now an explicit expression for the response function,* namely its *spectral representation*:

$$R_{pqp'q'}(\omega) = \frac{1}{\hbar}\sum_{\nu>0}\left(\frac{\langle 0|a_q^+ a_p|\nu\rangle\langle\nu|a_{p'}^+ a_{q'}|0\rangle}{\omega - \Omega_\nu + i\eta} - \frac{\langle 0|a_{p'}^+ a_{q'}|\nu\rangle\langle\nu|a_q^+ a_p|0\rangle}{\omega + \Omega_\nu + i\eta}\right). \tag{8.130}$$

Again, the index pairs pq and $p'q'$ run only over *ph* and *hp* pairs. All other matrix elements of R vanish in RPA order. The form (8.130), however is *more general*. If we use exact eigenfunctions $|\nu\rangle$ and exact energies $\hbar\Omega_\nu$ of the system, $R_{pq\,p'q'}(\omega)$ in Eq. (8.130) is just the exact response function. This can easily be seen by using time-dependent perturbation theory (see [No 64a, Chap. 2]) for the calculation of the change in the exact wave function produced by the external field F:

$$|\Psi(t)\rangle = |0\rangle + \sum_{\nu>0}|\nu\rangle\left\{\frac{\langle\nu|F|0\rangle}{\hbar(\omega - \Omega_\nu) + i\eta}e^{-i\omega t} - \frac{\langle\nu|F^+|0\rangle}{\hbar(\omega + \Omega_\nu) - i\eta}e^{i\omega t}\right\}.$$

The states $|0\rangle$ and $|\nu\rangle$ are the stationary eigenstates of the system without perturbation. The transition density corresponding to this wave function $\rho^{(1)}$ is then given by Eq. (8.125) with the exact response function (8.130).

If we introduce the response function R^0 of the free system (without residual interaction $\tilde{v}$ in (8.124)),

$$R^0_{pq\,p'q'}(\omega) = \frac{\rho_q^{(0)} - \rho_p^{(0)}}{\hbar\omega - \epsilon_p + \epsilon_q + i\eta}\delta_{pp'}\delta_{qq'},$$

* For the sake of completeness we have added the infinitesimal positive parameter η, which determines the boundary condition in the case of unbound states (see Sec. 8.5.4). In a bound state problem η can be set equal to zero.

we can finally, in RPA approximation, derive another equation for $R(\omega)$, the so-called *linearized Bethe–Salpeter* equation (F.75), viz.

$$R_{pq\ p'q'} = R^0_{pq\ p'q'} + \sum_{\substack{p_1q_1\\p_2q_2}} R^0_{pq\ p_1q_1}\tilde{v}_{p_1q_2\,q_1p_2}R_{p_2q_2\ p'q'}. \tag{8.131}$$

The correctness of this equation can be verified simply by multiplying by $(\hbar\omega - \epsilon_p + \epsilon_q)$ and using the definition of R^0, the spectral representation (8.130) for R and the RPA equation (8.126).

8.5.2 Calculation of Excitation Probabilities and Schematic Model

Another useful property of the linear response function lies in the fact that its imaginary part is related to the total transition probability (8.72). We define

$$R_F(\omega) := \mathrm{Tr}\big(f^+\rho^{(1)}(\omega)\big) = \sum_{pqp'q'} f^*_{pq}R^{(\omega)}_{pqp'q'}f_{p'q'} \tag{8.132}$$

and use the relation $1/(\omega + i\eta) = P(1/\omega) - i\pi\delta(\omega)$ in Eq. (8.130) to obtain

$$ImR_F(\omega) = -\pi\sum_{\nu>0}|\langle\nu|F|0\rangle|^2\delta(\hbar\omega - \hbar\Omega_\nu), \qquad \omega > 0. \tag{8.133}$$

We get the energy-weighted sum rule (8.154) by integrating this function,

$$-\frac{\hbar^2}{\pi}\int_0^\infty \omega ImR(\omega)\,d\omega = \sum_\nu \hbar\Omega_\nu|\langle\nu|F|0\rangle|^2 \equiv S_1, \tag{8.134}$$

and the transition matrix element $|\langle\nu|F|0\rangle|^2$ as the residue of $R_F(\omega)$ at the pole $\omega = \Omega_\nu$.

The solution becomes extremely simple in the schematic model, where the separable ansatz for the ground state correlation matrix element is now:

$$\tilde{v}_{mjin} = \lambda D_{mi}D^*_{nj}, \qquad \tilde{v}_{mnij} = \lambda D_{mi}D_{nj},$$

where D is identical with the external field operator F. From the Bethe–Salpeter equation (8.131), we get

$$R_D(\omega) = R^0_D(\omega)[1 + \lambda R_D(\omega)]$$

with

$$R^0_D(\omega) = \sum_{pqp'q'} D^*_{pq}R^0_{pqp'q'}D_{p'q'} = \sum_{mi}|D_{mi}|^2\left(\frac{1}{\hbar\omega - \epsilon_m + \epsilon_i + i\eta} - \frac{1}{\hbar\omega + \epsilon_m - \epsilon_i + i\eta}\right).$$

Solving for $R_D(\omega)$ yields

$$R_D(\omega) = \frac{R^0_D}{1 - \lambda R^0_D}.$$

The poles of $R_D(\omega)$ give the excitation energies Ω_ν and thus in the schematic model

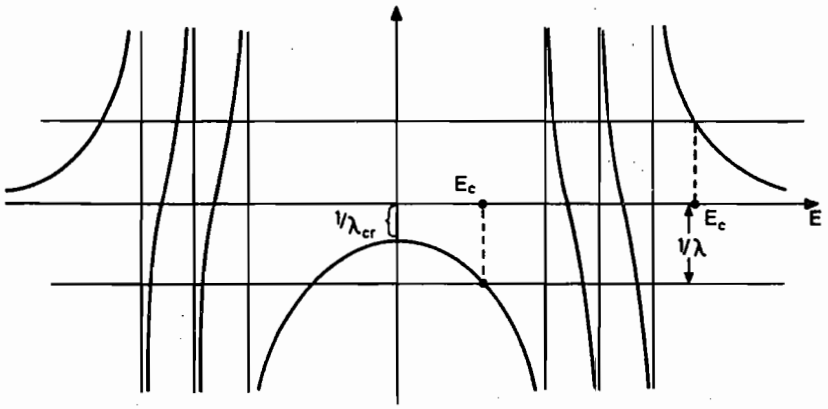


Figure 8.11. Graphical solution of the dispersion relation (8.135).

we have the following dispersion relation.

$$\frac{1}{\lambda} = R_D^0(\Omega_\nu) = \sum_{mi} |D_{mi}|^2 \frac{2\epsilon_{mi}}{\hbar^2\Omega_\nu^2 - \epsilon_{mi}^2}; \qquad \epsilon_{mi} = \epsilon_m - \epsilon_i. \tag{8.135}$$

(We can drop the infinitessimal η because we are only interested in bound states for which $\Omega_\nu \neq \epsilon_{mi}$.)

Equation (8.135) can be solved graphically as in the TDA case. Comparing Fig. 8.11 with Fig. 8.4, we notice two qualitative differences which are due to the RPA ground state correlations:

(i) In the case where the residual interaction becomes stronger than the critical value (i.e., $\lambda < \lambda_{\text{crit}}$) the energy of the low-lying collective state becomes imaginary.
(ii) The ($T=0$)-RPA state is shifted further down than its corresponding TDA state for a comparable interaction strength λ.

We can study this more closely in the degenerate case. If we put all ϵ_{mi} equal to ϵ we have, from Eq. (8.135),

$$E_{\text{coll}}^2 = \epsilon^2 + 2\epsilon\lambda \sum_{mi} |D_{mi}|^2.$$

In the degenerate case, therefore, we have for λ_{crit}

$$\lambda_{\text{crit}} = -\frac{\epsilon}{2\sum_{mi} |D_{mi}|^2}.$$

This is the point where the chosen HF-basis no longer gives the minimum for the ground state energy. The true minimum now occurs in a different HF-solution, which turns out to be deformed. We therefore often call λ_{crit} the point at which a phase transition from spherical state into a deformed shape of the nucleus occurs (for example, $D \sim r^2 Y_{20}$). In a comparison with the TDA case, we see that

$$E_{\text{coll}}^{\text{TDA}}(\lambda_{\text{crit}}) = \frac{\epsilon}{2}$$

and

$$E_{\text{coll}}^{\text{TDA}}(2\lambda_{\text{crit}}) = 0,$$

that is, in the schematic model the "phase transition" occurs in the TDA case at only twice the interaction strength.

We now want to turn to the calculation of transition matrix elements. We have to calculate the residue of $R_D(\omega)$ at the pole $\omega=\Omega_\nu$. In the vicinity of Ω_ν, $R_D(\omega)$ has the form:

$$R_D(\omega)=\frac{R_D^0(\Omega_\nu)}{-\lambda(dR^0/d\omega)|_{\omega=\Omega_\nu}\cdot(\omega-\Omega_\nu)}\,.$$

We therefore get

$$|\langle\nu|D|0\rangle|^2=-\frac{\hbar}{\lambda^2}\left(\frac{dR^0}{d\omega}\right)^{-1}_{\Omega_\nu}=\left(\lambda^2\sum_{mi}|D_{mi}|^2\frac{4\epsilon_{mi}\hbar\Omega_\nu}{\left(\hbar^2\Omega_\nu^2-\epsilon_{mi}^2\right)^2}\right)^{-1}.$$

In the degenerate case this yields

$$|\langle\nu|D|0\rangle|^2=\frac{\epsilon}{E_{\text{coll}}}\sum_{mi}|D_{mi}|^2.$$

We see that for the low-lying states the transition probability is enhanced as compared to the TDA value (8.21) by a factor ϵ/E_{coll}. For the collective octupole and quadrupole states this factor can be as large as two. Similar but less pronounced results are found in realistic calculations (see Sec. 8.6).

8.5.3 The Static Polarizability and the Moment of Inertia

We can apply linear response theory to calculate the change of the nuclear energy and the nuclear deformation in a static external field $-\lambda F$. This corresponds to the solution of the constrained HF equation

$$[h[\rho]-\lambda f,\rho]=0 \tag{8.136}$$

in linear response theory. For the change of the density, we get from Eq. (8.118) and (8.125),

$$\rho^{(1)}=-\lambda R(\omega=0)f=\lambda\mathcal{S}^{-1}f. \tag{8.137}$$

The static polarizability α is defined by the change of the expectation value of F:

$$\langle\lambda|F|\lambda\rangle=\langle 0|F|0\rangle+\lambda\cdot\alpha; \tag{8.138}$$

and the change in energy is given by

$$\langle\lambda|H|\lambda\rangle=\langle 0|H|0\rangle+\tfrac{1}{2}\lambda^2\cdot\alpha. \tag{8.139}$$

The fact that we have the same constant α in both cases is called *Feynman's theorem* [Fe 39] and can be derived in analogy to Eq. (3.81) for density dependent forces also.

Finally, for the curvature of the energy function at the stationary point $\lambda=0$, which is equivalent to the static polarizability, the compression

modulus [BGG 76], or the spring constant [MP 73], we get

$$\alpha = \frac{d^2E}{d\lambda^2} = \frac{d}{d\lambda}\langle\lambda|F|\lambda\rangle = \frac{d}{d\lambda}\mathrm{Tr}(f^+\rho^{(1)}) = f^+\mathbb{S}^{-1}f. \tag{8.140}$$

From Eq. (8.132) we have

$$\alpha = -R_F(\omega=0) = 2\sum_{\nu>0}\frac{|\langle\nu|F|0\rangle|^2}{E_\nu - E_0}, \tag{8.141}$$

which shows that the static polarizability is closely connected to the sum rule S_{-1} [see Eq. (8.150)]. Using the exact states $|\nu\rangle$ and exact energies we obtain Eq. (8.141) by calculating the exact polarizability α_{exact} from first-order perturbation theory. The derivation shows that we get the polarizability in HF-approximation α_{HF} by using RPA wave functions and energies on the r.h.s. of Eq. (8.141). This can also be generalized to the dynamic polarizability [SLO 78].

The calculation of the *moment of inertia* within the self-consistent cranking theory (see Sec. 7.7) provides a simple application of these considerations. In this case, the external field λF is given by the cranking operator ωJ_x. An exact solution of Eq. (8.136) has been discussed in Section 7.7. Thouless and Valatin [TV 62] proposed to solve it in linear response theory. In this case, from Eq. (8.123) we get the Thouless–Valatin equation:

$$\mathbb{S}\rho^{(1)} = \omega J_x,$$

which is equivalent to Eq. (8.106). As discussed there, we have to be a little careful with the solution, since in this case $\mathbb{S}$ is a singular matrix. Nevertheless, we get the result for the moment of inertia from

$$\vartheta_{\text{TV}} = \left.\frac{d^2E}{d\omega^2}\right|_{\omega=0} = 2\sum_{\nu>0}\frac{|\langle\nu|J_x|0\rangle|^2}{E_\nu - E_0}, \tag{8.142}$$

where the sum over υ does not include the spurious state. This is a very general expression. It contains, as special cases, the Inglis formula (3.89) ($|\nu\rangle$ pure *ph* states) and the Belyaev formula (3.93) ($|\nu\rangle$ pure 2-quasiparticle states). But it also contains the more general case in which pairing correlations are included into the RPA theory (see Sec. 8.9). In this case, $|\nu\rangle$ are solutions of the quasi-particle RPA equations.

8.5.4 RPA Equations in the Continuum

Up to now we have always assumed a discrete spectrum of excited states $|\nu\rangle$. In fact, only the low-lying excited states are really discrete levels. All higher excitations lie in the continuum, that is, they are resonances. As discussed in Section 8.3, they are observed as maxima in the cross section of a suitable reaction.

The simplest way to overcome this problem is to work in a truncated oscillator basis. In such a basis there are only discrete levels. Single-particle resonances of the HF potential are approximated by bound single-particle states. Since the residual

interaction has a short range, and since the matrix elements entering the RPA matrix always contain two hole indices which correspond to exponentially decreasing single-particle wave functions, we need the wave functions of the unbound particle states only in the region of the nucleus, and we can approximate them there by a bound state. As a result of such a discretization, we get no giant resonances from the solution of the RPA equations (8.70), but groups of discrete states in this region (see Fig. 8.12) which show an enhanced strength $|\langle\nu|F|0\rangle|^2$ for the appropriate transition operator (as, for instance, $F=r^{\lambda}Y_{\lambda\mu}$ in the case of surface vibrations of multipolarity λ).

This method, however, provides only an approximative solution to the problem which becomes rather poor in cases where the single-particle resonances are spread out over a wide energy range. In reality, the problem is not a bound state problem. There is a solution for each energy ω, and the interesting quantity is now the excitation strength

$$S(\omega)=\sum_{\nu>0}|\langle\nu|F|0\rangle|^2\delta(\hbar\omega-\hbar\Omega_{\nu}), \tag{8.143}$$

which shows poles of discrete values of ν (bound states) and has a resonance behavior for continuous quantum numbers ν. In fact we have already seen in Eq. (8.38) that for the example of the giant dipole resonance this is exactly the measurable quantity. From Eq. (8.133) we get:

$$S(\omega)=-\frac{1}{\pi}ImR_F(\omega).$$

Therefore, we have to calculate the response function $R_{pqp'q'}(\omega)$ of Eq. (8.130).

It turns out that the width of the giant resonances has two origins:

(i) The wave function of the giant resonance is given as a superposition of *ph* states, where the *p*-state is a scattering solution of the single-particle HF potential. This means that there is always a certain probability that the state will decay into a free particle and a hole state in the $A-1$ nucleus. This part of the width is called the *decay width* [MW 69a]. We are able, as we shall see in a moment, to describe this if we use scattering solutions of the single-particle HF equations in the calculation of $R(\omega)$.

(ii) There are more complicated states, for instance, $2p-2h$ excitations, which can also contribute to the wave function. They are neglected in the RPA approximation. It means that in reality a *ph*-component of the wave function is spread out over many $2p-2h$ components. Therefore this part is called the spreading width. We are not able to calculate this in the present approximation (see for example [WS 72, HA 76, SSV 77, DKS 77, AY 78, BBB 79b].)

Several authors have treated the continuum within the TDA or the RPA [BG 65, DH 68, LV 68]. Because of the continuous variable ϵ of the single-particle energy in the continuum, we find a set of integrodifferential equations that are similar to the continuum shell model equations [We 67, MW 69a]. Their solution requires a big numerical effort [RMS 67, BH 67, Bi 72, KKS 77]. We do not want to present this matter further here but will instead discuss a method for calculating $R(\omega)$ for the case of zero range forces.

In cases in which the residual interaction consists only of a δ-force or derivatives of it [such as the Skyrme force (4.104)ff], Bertsch et al. [Be 73,

SB 75, BT 75, LG 76b, Ts 78] have developed a method of solving the Bethe–Salpeter equation (8.131) for the response function R in coordinate space. We present it here for the simple case of a pure δ-force $v(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \mathbf{r}_4) = V_0 \cdot \delta(\mathbf{r}_1 - \mathbf{r}_2)\delta(\mathbf{r}_2 - \mathbf{r}_3)\delta(\mathbf{r}_3 - \mathbf{r}_4)$ and neglect spin variables.

Using Eq. (8.131) and transforming it into coordinate space, we get, for the shell model response function R^0,

$$R^0(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}'_1, \mathbf{r}'_2, \omega) = \sum_{mi} \frac{\varphi_m(\mathbf{r}_1)\varphi_i^*(\mathbf{r}_2)\varphi_m^*(\mathbf{r}'_1)\varphi_i(\mathbf{r}'_2)}{\hbar\omega - \epsilon_m + \epsilon_i + i\eta} - \frac{\varphi_i(\mathbf{r}_1)\varphi_m^*(\mathbf{r}_2)\varphi_i^*(\mathbf{r}'_1)\varphi_m(\mathbf{r}'_2)}{\hbar\omega + \epsilon_m - \epsilon_i + i\eta}, \tag{8.144}$$

where $\varphi_q(\mathbf{r})$ are eigenfunctions of the single-particle HF Hamiltonian h^0 with eigenvalue ϵ_q. Because of the δ-character of the force, we will only need the function R^0 for $\mathbf{r}_1 = \mathbf{r}_2$ and $\mathbf{r}'_1 = \mathbf{r}'_2$:

$$R^0(\mathbf{r}, \mathbf{r}', \omega) = \sum_i \left\{ \varphi_i^*(\mathbf{r}) \langle \mathbf{r} | \frac{1}{\hbar\omega + \epsilon_i + i\eta - h^0} | \mathbf{r}' \rangle \varphi_i(\mathbf{r}') + \varphi_i^*(\mathbf{r}') \langle \mathbf{r}' | \frac{1}{-\hbar\omega + \epsilon_i + i\eta - h^0} | \mathbf{r} \rangle \varphi_i(\mathbf{r}) \right\}. \tag{8.145}$$

To see that (8.145) follows from (8.144), we must insert a complete set $\sum_s |s\rangle\langle s|$ and realize that the hole contributions vanish.

In the case of a pure δ-force, h^0 contains the kinetic energy and a local one-body potential. We can apply scattering theory in a one-body potential (see [Me 61, Vol. II, Chap. XIX.13]) and derive for a given partial wave the radial part of the single-particle Green's function viz:

$$G(r, r', \omega) \equiv \langle r | \frac{1}{\hbar\omega + i\eta - h^0} | r' \rangle = \frac{2m}{\hbar^2} \frac{v(r_>)w(r_<)}{W(v, w)}. \tag{8.146}$$

Here $r_>$ and $r_<$ denote the greater and the lesser of r and r' and v and w are two linear independent solutions of $(h^0 - \omega)v = 0$ with the boundary conditions:

$r = 0$: $v(r)$ regular, $w(r)$ irregular;

$r \to \infty$ $\omega < 0$, $v(r)$ increases and $w(r)$ decreases exponentially;

$\omega > 0$, $v(r)$ standing wave $w(r)$ outgoing wave.

Finally, $W(v, w)$ is the Wronskian of v and w.

Having a solution of the HF problem, we know the radial part of the bound state wave function $\varphi_i(r)$, and the functions v and w at a set of mesh points in coordinate space. For each ω, we can therefore get the function $R^0(\mathbf{r}, \mathbf{r}', \omega)$ from Eq. (8.145) and (8.146) with the aid of some angular momentum coupling.

For the local response function $R(\mathbf{r}, \mathbf{r}') = R(\mathbf{r}\mathbf{r}, \mathbf{r}'\mathbf{r}')$ with a pure δ-force, we get from (8.131)

$$R(\mathbf{r}, \mathbf{r}', \omega) = R^0(\mathbf{r}, \mathbf{r}', \omega) + V_0 \int d^3r'' \, R^0(\mathbf{r}, \mathbf{r}'', \omega) R(\mathbf{r}'', \mathbf{r}', \omega). \tag{8.147}$$

Using angular momentum coupling techniques [BT 75], the three-dimensional integral can be reduced to a one-dimensional integral over the radial coordinate. Introducing mesh points in r-space, we find an inhomogeneous linear matrix equation whose solution gives $R(\mathbf{r}, \mathbf{r}')$.

For a local operator $F(\mathbf{r})$ (for instance, a multipole operator), we are now able to calculate the transition density $\rho^{(1)}(r)$ from Eq. (8.125), which is sometimes also called the *form factor*:

$$\rho^{(1)}(\mathbf{r}, \omega) = \int d^3r' \, R(\mathbf{r}, \mathbf{r}', \omega) F(\mathbf{r}'); \tag{8.148}$$

and the strength function (8.142)

$$S(\omega) = -\frac{1}{\pi} Im \int d^3r \, d^3r' \, F^*(\mathbf{r}) R(\mathbf{r}, \mathbf{r}', \omega) F(\mathbf{r}'). \tag{8.149}$$

(An example of such a calculation is given in Fig. (8.13).)

8.6 Applications and Comparison with Experiment

Over the years, there has been an enormous number of applications of the TDA and RPA approximations in nuclei starting with simple configuration mixing of a few *ph* levels in O^{16} up to the extended RPA calculations with inclusion of the continuum in heavy nuclei. It goes far beyond the scope of this book to give a review of this work. We wish only to pick a few examples to demonstrate certain essential points. We therefore restrict ourselves to spherical closed-shell nuclei such as ^{16}O, ^{40}Ca, ^{50}Zr, and ^{208}Pb. In a spherical basis, we can use angular momentum coupling techniques [Ed 57] to diagonalize the Hamiltonian in the eigenspaces of angular momentum and parity. This fact reduces the dimension of the corresponding matrices enormously. In $N = Z$ nuclei, we can furthermore neglect the Coulomb force to a good approximation and use the isospin symmetry.

8.6.1 Particle–Hole Calculations in a Phenomenological Basis

In most of the calculations the self-consistent HF potential has been replaced by a phenomenological shell model potential, such as an oscillator potential whose oscillator length was adjusted to the empirical rms radius. This potential provides single-particle wave functions for the calculation of the *ph*-matrix elements (8.71) of a residual interaction. Since the final energy levels—at least for the non-collective states—depend very sensitively on the particle–hole energies $\epsilon_m - \epsilon_i$, in fact, usually not the single-particle energies of the phenomenological potential have been used, but instead taken the measured single-particle spectrum energies determined from neighboring odd nuclei

$$\epsilon_m = E(m, A+1) - E(0, A),$$
$$\epsilon_i = E(0, A) - E(i, A-1).$$

One of the oldest calculations of this type is the TDA calculation of Elliot and Flowers [EF 57] for the negative parity levels in ^{16}O. They used one shell below the Fermi surface ($1p$) and one shell above the Fermi surface ($2s - 1d$ shell). This gives

five *ph*-states for the 1^- channel and three *ph* states for the 3^- channel. With a Yukawa interaction and a Rosenfeld exchange mixture [EF 57, Table 4.3], they found a highly collective $3^-T=0$ state at $\simeq 7$ MeV and two collective $1^-T=1$ states at 22.6 and 25.2 MeV. The octupole state exhausted 68% and the two dipole states exhausted 97% of the S_0 sum rule in this configuration space. This agrees qualitatively with the results of the schematic model (Sec. 8.2.2). The strength is concentrated mainly in one level shifted away from the unperturbed single-particle energies (~ 11.5 MeV).

The position of the energy levels is in good agreement with experiment—the two dipole states, which are discrete levels in this calculation, correspond to the two peaks in the dipole-resonance (Fig. 8.1). The absolute value of the excitation probability of the 3^- state is, however, a factor of 3–6 too small. One reason for this is the neglect of ground state correlations in the TDA; another reason is that the configuration space which includes only $1\hbar\omega_0$ excitations is probably too small (see below).

Gillet et al. [Gi 64, GV 64, GS 64, GGS 66] investigated the influence of ground state correlation by comparing TDA with RPA calculations in the closed shell nuclei ^{12}C, ^{16}O, ^{40}Ca, and ^{208}Pb. They again used a harmonic oscillator basis and a Gaussian force [Eq. (4.100)]. The parameters of the exchange mixture (Tab. (4.3)) were adjusted by extensive least-square fits to the ^{12}C and ^{16}O data. They found that the RPA leaves the TDA more or less unchanged for $T=1$ levels. The low-lying $T=0$ states, however, were reproduced better by the RPA. In particular, the RPA excitation probabilities were considerably larger than the TDA ones, although actually not reaching the experimental value. The reason is again the size of the configuration space, which included only $1\hbar\omega_0$ excitations. This fact in particular causes very poor results for positive parity states.

In detail, the results depend very much on the residual interaction. As we have already seen in Chapter 4, there have been attempts to derive such an interaction from the bare nucleon force. It is therefore very interesting to apply such forces to RPA calculations also. In this case, we have no free parameter and cannot necessarily expect the same agreement with the experimental data as with a phenomenologically adjusted force. Several groups [MMG 67, BK 69, KBB 70] have used the Kuo–Brown effective interaction [KB 66]. As discussed in Chapter 4, there are still many open problems inherent in the derivation of this force, and it is not absolutely clear how to modify an effective interaction originally deduced for *pp*-calculations for a *ph* theory. Nevertheless, the results are encouraging. Many of the collective energies are reproduced quite well. It turns out that the screening term (Fig. 4.7) plays an important role. Without it, the energy of the collective 3^- state in ^{208}Pb becomes imaginary [KBB 70]. The calculations have again been carried out in an oscillator basis and the configuration space includes only $1\hbar\omega_0$ excitations. Therefore, the absolute strength of the excitation probabilities is again too small.

Ring and Speth [RS 74a] used the theory of finite Fermi systems developed by Migdal [Mi 67]. This theory starts from quite a different point of view. It describes the interaction between quasi-particles and quasi-holes by a renormalized effective interaction which is short ranged and density dependent (Eq. (4.113)) and ends up with the RPA equations for the calculation of collective states. The force is phenomenological in as far as its parameters have to be adjusted to experimental values (such as energies and transition probabilities of collective states in even nuclei and electromagnetic moments and transitions probabilities in neighboring odd nuclei (see Sec. 9.3.5). This has been done in the lead region [RBS 73, BSK 73,

NW 74, BER 75, SWW 77] and the force has been applied in extensive RPA calculations in ^{208}Pb [RS 73, RS 74a, ERW 78] and other doubly closed shell nuclei [KS 74].

These calculations are different from the work discussed above in so far as they use (i) a Woods–Saxon basis, which is particularly important and more realistic than the harmonic oscillator for heavy nuclei, (ii) a large configuration space: two shells below and two shells above the Fermi surface, and (iii) a density-dependent interaction, which is slightly repulsive in the nuclear interior and attractive outside. The results are very encouraging. With six parameters adjusted once and for all, it is possible to reproduce nearly all of the experimental measured effective charges in

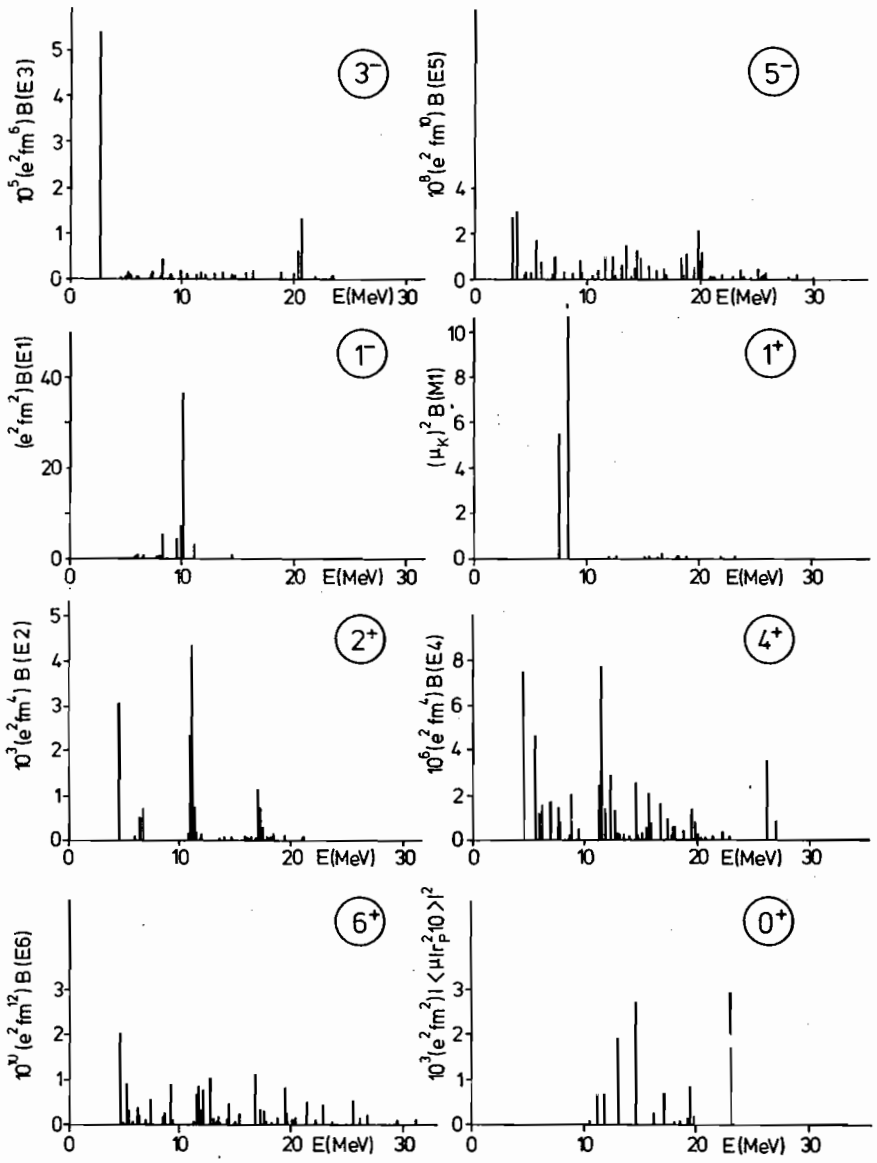


Figure 8.12. Distribution of multipole strength in ^{208}Pb (from [RS 74a]).

odd nuclei as well as the energies and the excitation probabilities of the excited states in even nuclei (see Table 8.1). Since the configuration space includes $2\hbar\omega$ excitations it is now also possible to reproduce positive parity states.

These calculations have been carried out in a discretized basis (the Woods–Saxon potential is expanded in a harmonic oscillator basis). The giant resonance therefore consists of discrete lines. Figure 8.12 shows the distribution of the strength. In particular, it shows the low-lying 3^-, 2^+, 4^+, 6^- collective states, but also the giant resonances, such as 1^- (which lies too low in energy), 2^+ (which is split into a $T=0$ part and a $T=1$ part), a breathing mode (0^+) at $\sim$14 MeV, and higher resonances. To get a continuous resonance an RPA must be calculated fully in the continuum, as has been discussed in Sec. 8.5.4 [KBF 74, TW 78].

Similar calculations in a discretized basis have been carried out with a momentum dependent residual interaction [KH 76].

Table 8.1. Energies and excitation probabilities of low-lying collective states in ^{208}Pb, calculated with the Migdal force in RPA theory [RS 74a].

	$E(I)$ [MeV]		$B\binom{E}{M}I$ $[e^2\,\mathrm{fm}^{2I}]$ $[\mu^2\,\mathrm{fm}^{2I-2}]$	
	theory	exp	theory	exp
3^-	2.63	2.61	546	$540 \pm 30 \times 10^3$
5^-	3.39	3.19	285	$462 \pm 55 \times 10^6$
5^-	3.82	3.71	301	330×10^6
2^+	4.49	4.07	3070	2965
4^+	4.69	4.32	757	1287×10^4
6^+	4.77	4.42	210	230×10^8
1^+	7.50	7.56	568	5.9
1^+	8.30	7.99	11.00	10.0
2^-	7.51		11.00×10^3	

8.6.2 Particle–Hole Calculations in a Self-Consistent Basis

As we have seen, RPA calculations with phenomenological density-dependent forces are able to describe many of the excited states in spherical closed shell nuclei. In Chapter 5 we learned that ground state properties are reproduced rather well in HF theory with Skyrme forces, and in Section 8.5 we constructed a method of deriving RPA theory from the density dependent HF theory: we have only to use the second derivative of the total energy with respect to the density as the *ph*-force [see Eq. (8.124)]. In the last few years this concept has been used by several groups [BT 75, LB 76, BGG 76, KKS 77, BG 77].

Starting from an HF calculation in coordinate space they either diagonalize the resulting HF single-particle Hamiltonian in an oscillator basis with nine or ten major shells to obtain discrete single-particle levels for a discretized RPA calculation or use the HF wave function and calculate in coordinate space. In the first case, we get discrete eigenvalues of the RPA equation, in particular the low-lying bound states and discretized resonances. In the other case, the continuum is taken

fully into account and we immediately get the strength function $S(\omega)$ [Eq. (8.149)], which has poles at the bound states.

The result of this calculation is that the usual Skyrme force—in its different versions [BFG 75]—which reproduces ground state properties very nicely, is not able to describe all the excited states [SLR 73, SZ 73, Ch 75, BJS 75]. This fact is not very disturbing, as the Skyrme force was adjusted so as to give the right energy and the right density at the equilibrium point of the functional $E[\rho]$. In particular, this means that the minimum is at the right value ρ_0 and has the right energy $E[\rho_0]$. However, it is not necessary that the curvature of the energy surface, $\delta^2 E/\delta\rho^2$, which gives the *ph* force, also be reproduced.

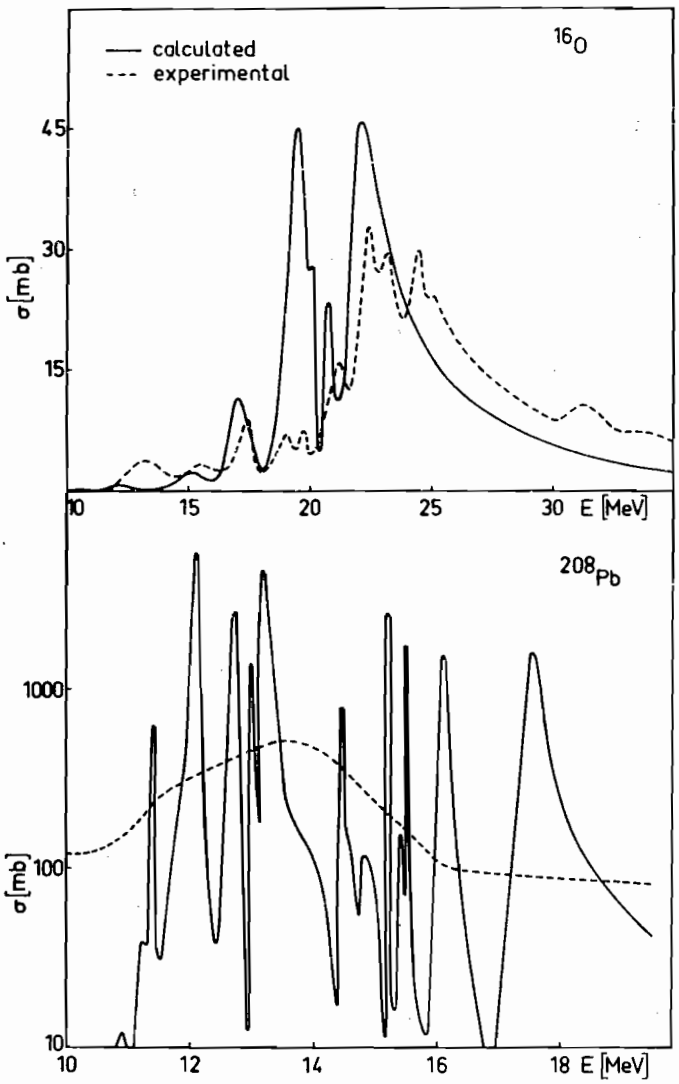


Figure 8.13. The photo cross section of the giant dipole resonance in ^{16}O and ^{208}Pb calculated with a continuum RPA calculation. The dashed lines correspond to experimental values. (From [LG 76b].)

The different groups, therefore, proposed generalizations of the Skyrme force that will not spoil their properties in HF calculations, such as a momentum-dependent three-body term [LB 76] $k^2 \cdot \delta(\mathbf{r}_{12})\delta(\mathbf{r}_{13})$ or density dependence of the momentum dependent S- and P-wave interaction [KKS 77]. In this way, one is able to reproduce the low-lying bound states more or less correctly and, in general, gets the right positions of the giant resonances. However, one is not able—even with the continuum calculations—to reproduce the proper width and the fine structure of the giant resonances. The reason is that only the decay width has been calculated (see Sec. 8.5.4) and not the spreading width. To do this we must include higher configurations such as $2p-2h$ or $3h-3h$ states [WS 72]. This goes beyond the RPA approach. In Fig. 8.13, we observe that in ^{208}Pb, in particular, the theoretical curve for the giant resonance is split into many sharp peaks and coupling to higher configurations should wash this out.

8.7 Sum Rules

We have already seen that in the calculation of the total cross-section of the giant dipole resonance in Section (8.3.3), the sum rules form a very important tool in the theory of collective excitations. In many cases, they allow a calculation of global properties in a simple way and are therefore useful in testing different approximation schemes. Sum rules give less detailed information than the RPA solutions, but they are much easier to calculate and easier to interpret [No 78, BLM 79]. More particularly, they allow connections to be made to the older macroscopic or hydrodynamical models.

8.7.1 Sum Rules as Energy Weighted Moments of the Strength Functions

In general, a sum rule is related to a Hermitian single-particle operator $F=\sum f_{pq}a_p^+a_q$ and given by

$$S_k \equiv \sum_\nu (E_\nu - E_0)^k |\langle \nu|F|0\rangle|^2. \tag{8.150}$$

It gives the kth moment of the distribution of the excitation strength produced by the one-body operator F. The $|\nu\rangle$s represent the complete set of eigenstates of the exact Hamiltonian H with the energies E_ν. Using the completeness relation, we find

$$S_k = \langle 0|F(H-E_0)^k F|0\rangle. \tag{8.151}$$

We will see that in some cases this expression can be calculated in a rather simple way.

Following Lane et al. [MBF 76, BML 76, BLM 79], we can define a set

of energies

$$\mathcal{E}_k = \left(\frac{S_k}{S_{k-2}}\right)^{1/2} \tag{8.152}$$

which characterizes the strength distribution (8.150). If it is sharply peaked at a certain energy, then all the $\mathcal{E}_k$ coincide. The degree to which they are different reflects the width of the distribution.

Using the Schwarz inequality for the moments $S_{k+2} \cdot S_k \geqslant S_{k+1} \cdot S_{k+1}$, we deduce [La 75] that

$$\mathcal{E}_{-1} \leqslant \mathcal{E}_0 \leqslant \mathcal{E}_1 \leqslant \bar{E} \equiv \frac{S_1}{S_0} \leqslant \mathcal{E}_2 \leqslant \cdots$$

and get for the mean square fluctuation about the mean energy [BLM 79]

$$\rho^2 \equiv \mathcal{E}_2^2 - \bar{E}^2 \leqslant \tfrac{1}{4}(\mathcal{E}_3^2 - \mathcal{E}_1^2). \tag{8.153}$$

As we shall see, we can evaluate the moments S_{-1}, S_1, and S_3 rather generally. Therefore, in this way we can gain information about the energy of a resonance and its width.

8.7.2 The S_1-Sum Rule and the RPA Approach

The most important sum rule is the energy weighted sum rule S_1. It can be written as a double commutator [see Eq. (8.42)]

$$S_1 = \sum_\nu (E_\nu - E_0)|\langle \nu|F|0\rangle|^2 = \tfrac{1}{2}\langle 0|[F,[H,F]]|0\rangle. \tag{8.154}$$

This relation holds for a set of exact eigenstates $|\nu\rangle$ of H. We usually have only approximative states $|\nu\rangle$ and approximative energies E_ν. It is a test of the validity of any approximation to see whether it fulfills the sum rules.

Thouless [Th 61a] showed that the sum rule (8.154) is satisfied if the left-hand side is evaluated with RPA wave functions and energies and the right-hand side is calculated using the HF ground state wave function. In the following we give a proof of this theorem for the case of Hermitian one-body operators F and G [La 75]. Using the notation of Eq. (8.72), we find for density independent forces with (8.76) and (8.79)

$$\begin{aligned}\langle \mathrm{HF}|[F,[H,G]]|\mathrm{HF}\rangle &= (F^* - F)\begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix}\begin{pmatrix} G \\ -G^* \end{pmatrix} \\ &= f^+ \mathcal{X}\Omega\mathcal{R}\mathcal{X}^+ g.\end{aligned} \tag{8.155}$$

Finally, expression (8.72) for the strengths $\langle 0|F|\nu\rangle$ yields

$$\langle HF|[F,[H,G]]|HF\rangle = \sum_\nu \hbar\Omega_\nu(\langle 0|F|\nu\rangle\langle\nu|G|0\rangle + \langle 0|G|\nu\rangle\langle\nu|F|0\rangle). \tag{8.156}$$

It is easy to generalize this relation [GLM 78]:

$$f^{+}\begin{pmatrix} A & B \\ -B^{*} & -A^{*} \end{pmatrix}^{n} \mathcal{R}g = \sum_{\nu} (\hbar\Omega_{\nu})^{n} \{\langle 0|F|\nu\rangle\langle\nu|G|0\rangle + (-)^{n+1}\,\text{c.c.}\}. \tag{8.157}$$

In the case of *density dependent* forces, however, the matrices A and B are defined by Eq. (8.124), and this proof does not apply.

In fact, in general we get additional terms on the rhs of Eq. (8.154). Nevertheless, it can be shown that the theorem also applies to local operators $F(\mathbf{r})$ and interactions whose density-dependent parts do commute with F [BG 77], as is the case for the usual Skyrme force (4.104).

From Eq. (8.155) we also recognize immediately that the sum rule S_1 is violated in the TDA approach, which is obtained by simply setting $B=0$, and conversely the TDA approximation conserves the sum rule S_0, which is violated by the RPA. The pure shell model without residual interaction fulfills both sum rules S_0 and S_1.

The sum rule S_1 is of special importance because it can be reduced to an expectation value of a one-body operator which can be evaluated rather easily, as will be seen in the following section.

8.7.3 Evaluation of the Sum Rules S_1, S_{-1}, and S_3

To evaluate the energy weighted sum rule, we shall restrict ourselves here to an isoscalar single-particle operator $F=\sum_i f(\mathbf{r}_i)$ and neglect momentum dependent parts in the residual interaction.*

In this case we have only a contribution coming from the kinetic energy, thus

$$S_1 = \frac{1}{2}\langle 0|[F,[H,F]]|0\rangle = \frac{\hbar^2}{2m} A\langle 0|(\nabla f)^2|0\rangle. \tag{8.158}$$

However, the dipole case is extremely simple, because F is linear and ∇f no longer depends on $\mathbf{r}$ (8.39). This part of the calculation is therefore model independent. In all other cases $(\nabla f)^2$ is still a single-particle operator which should not be too sensitive to correlations in the ground state $|0\rangle$. It can therefore be easily calculated within a model.

For surface vibrations we have already seen that $F=r^{\lambda}Y_{\lambda 0}$. Using the gradient formula [Ed 57], we may derive†

$$S_1(r^{\lambda}Y_{\lambda 0}) = \frac{\hbar^2}{2m} A\langle 0|(\nabla r^{\lambda}Y_{\lambda 0})^2|0\rangle = \frac{\hbar^2}{2m}\,\frac{(2\lambda+1)\lambda}{4\pi}\cdot A\cdot\langle r^{2\lambda-2}\rangle. \tag{8.159}$$

*We can show [BL 76] that Eq. (8.158) holds for the Skyrme force, too, because of the δ-character of the p^2-terms. This is, however, no longer true for isovector operators such as the dipole operator (8.39). We also gain contributions to the dipole sum rule from exchange forces, as discussed in Section 8.3.3 (see also [Ki 78]).

†For a nucleus with a constant density and a sharp surface we obtain:

$$\langle r^{L}\rangle = \frac{3}{L+3}R^{L}.$$

In the quadrupole case, S_1 is proportional to the mean square radius $\langle r^2 \rangle$. The same is true for $E0$ excitations—breathing modes—in which $F \propto r^2$ must be used (see [SF 74]).

We have already seen in Section 8.5.3 that the static polarizability α is proportional to the inverse energy-weighted sum rule S_{-1}; this fact can be used to calculate it in the RPA approximation using the solution of a constrained HF calculation $(H - \lambda F)$ in the limit of small λ-values (see also [SLO 78]).

$$S_{-1}^{\mathrm{RPA}} = \frac{1}{2} \frac{\partial^2}{\partial \lambda^2} \langle \Phi(\lambda) | H | \Phi(\lambda) \rangle_{\lambda=0}. \tag{8.160}$$

To calculate the sum rule S_3, we can use the identity [La 75, MBF 76, LOL 76]:

$$S_3 = \langle 0 | F (H - E_0)^3 F | 0 \rangle = -\frac{1}{2} \left(\frac{2\hbar^2}{m} \right)^2 \langle 0 | [G, [H, G]] | 0 \rangle \tag{8.161}$$

with $-(2\hbar^2/m)G = [H, F] = [T, F]$, which is a one-particle operator if the potential energy commutes with F. In this case, we can calculate the energy-weighted sum rule for the operator G. It can be obtained as the second derivative

$$S_3 = \frac{1}{2} \left(\frac{2\hbar^2}{m} \right)^2 \frac{\partial^2}{\partial \eta^2} E(\eta) \Big|_{\eta=0} \tag{8.162}$$

of the energy

$$E(\eta) = \langle 0 | e^{-\eta G} H e^{\eta G} | 0 \rangle. \tag{8.163}$$

This equation is also true if we replace the exact ground state $|0\rangle$ on the r.h.s. by the $|\mathrm{HF}\rangle$ ground state and S_3 by S_3^{RPA} on the left-hand side.

In the following, we shall restrict ourselves to isoscalar multipole vibrations ($f = r^\lambda Y_{\lambda 0}$) and breathing modes ($f = r^2$). Similar calculations have also been made for the isovector modes [BL 76].

For the operator G, we find

$$G = \frac{1}{2} (\nabla r^\lambda Y_{\lambda 0}) \cdot \nabla \ (\lambda \neq 0) \quad \text{and} \quad G = \frac{3}{2} + r \frac{\partial}{\partial r} \ (\lambda = 0), \tag{8.164}$$

where we have used $\Delta r^\lambda Y_{\lambda 0} = 0$. We see that $e^{\eta G}$ is, in fact, a *scaling operation*.

$$e^{\eta/2 (\nabla r^\lambda Y_{\lambda 0}) \nabla} \Phi(\mathbf{r}) \propto \Phi\left(\mathbf{r} + \frac{\eta}{2} \nabla r^\lambda Y_{\lambda 0} \right). \tag{8.165}$$

In the case $\lambda = 0$ and $\lambda = 2$, we get

$$e^{\eta G} \Phi(x_i, y_i, z_i) = \begin{cases} e^{3/2\eta} \Phi(x_i e^\eta, y_i e^\eta, z_i e^\eta), & (\lambda = 0), \\ \Phi(x_i e^{-\eta'}, y_i e^{-\eta'}, z_i e^{2\eta'}), & (\lambda = 2), \end{cases}$$

with $\eta' = \eta \cdot (5/16\pi)^{1/2}$. The factor $e^{3/2\eta}$ in the case $\lambda = 0$ shows the difference between the breathing modes, which change the density over the

whole volume, and the higher multipolarities, which involve only the nuclear surface. It is easy to calculate the contributions to $E(\eta)$ of the different parts of simple forces:

	$\lambda=0$	$\lambda=2$
Kinetic energy T	$e^{2\eta}\langle T\rangle$	$1/3(2e^{-2\eta'}+e^{+4\eta'})\langle T\rangle$
$V_0\propto\delta(\mathbf{r}-\mathbf{r}')$	$e^{3\eta}\langle V_0\rangle$	$\langle V_0\rangle$
$V_\rho\propto\rho\delta(\mathbf{r}-\mathbf{r}')$	$e^{6\eta}\langle V_\rho\rangle$	$\langle V_\rho\rangle$

and to derive the contributions to the sum rule S_3. As an example, we present the case of quadrupole vibrations ($\lambda=2$):

$$S_3(\lambda=2)=\frac{1}{2}\left(\frac{2\hbar^2}{m}\right)^2\frac{5}{16\pi}8\langle T\rangle.$$

Together with (8.159), we find for the energy $\mathcal{E}_3$ (8.152)

$$\mathcal{E}_3(\lambda=2)=\left(\frac{S_3}{S_1}\right)^{1/2}=\left(\frac{\hbar^2}{m}\frac{4\langle T\rangle}{A\langle r^2\rangle}\right)^{1/2}. \tag{8.166}$$

In the harmonic oscillator model, the expectation value of T is equal to the potential energy $\langle T\rangle=\frac{1}{2}m\omega_0^2\langle r^2\rangle A$. If we furthermore assume that all the strength is concentrated in a small energy interval, all the $\mathcal{E}_k$ should be roughly equal, and we then find for the energy of the giant quadrupole resonance ($T=0$) with $\hbar\omega_0\simeq 41A^{-1/3}$ [MeV],

$$E(2^+)=\sqrt{2}\,\hbar\omega_0\simeq 58\cdot A^{-1/3}\,[\mathrm{MeV}] \tag{8.167}$$

a formula which is in good agreement with experiment [Eq. (8.46)] and which has also been derived in a similar model by Mottelson and Suzuki [BM 75, Su 73]. It can be extended to higher isoscalar surface vibrations and gives

$$E(\lambda)=\sqrt{\lambda}\cdot 41\cdot A^{-1/3}\,[\mathrm{MeV}]\qquad(\lambda\geqslant 2). \tag{8.168}$$

Zamick [Za 73, Za 74a] has shown that the breathing mode is not well described by zero range forces which are unable to give saturation. One has to include finite range terms, or at least the p^2-terms, in the Skyrme force. With reasonable values for the force constants, we get

$$E(0^+)\simeq 2\hbar\omega_0\simeq 82A^{-1/3}\,[\mathrm{MeV}]. \tag{8.169}$$

We find that the energy of all the surface vibrations is $\propto A^{-1/3}$, which is in agreement with the experiment. From the simple hydrodynamical model, as discussed in Section 1.4, we find $E_\lambda=(C_\lambda/B_\lambda)^{1/2}\propto A^{-1/2}$. The reason for this discrepancy will be discussed in Section 13.3.

8.7.4 Sum Rules and Polarizabilities

In Eq. (8.160) we expressed the sum rule $S_{-1}(F)$, that is, the -1st moment of the strength distribution in a RPA calculation, by the RPA polarizability of the system in an external field F. This can be extended [GLM 78]. The RPA sum rules $S_{2n-1}(F)$ for arbitrary n can be expressed as the polarizability in a suitable single-particle field $F^{(n)}$.

We therefore start with the operator F and define

$$\begin{aligned} F^{(0)} &= \sum_{mi} f_{mi} a_m^+ a_i + f_{im} a_i^+ a_m \\ F^{(1)} &= i\big[H, F^{(0)}\big]' \\ F^{(2)} &= i\big[H, F^{(1)}\big]' \\ &\vdots \end{aligned} \tag{8.170}$$

where the prime means only the *ph* and *hp* part of the corresponding operator $F^{(n)}$. The $F^{(n)}$ is therefore a single-particle operator with vanishing *pp* and *hh* matrix elements.

Using the definitions (8.72) we get for the *ph* and *hp* matrix elements

$$f^{(n+1)} = i\mathbb{S}\mathcal{R} f^{(n)}. \tag{8.171}$$

With this relation we may also define the operators $F^{(n)}$ for negative values of n. From Eq. (8.157) we find that

$$\begin{aligned} S_{2n-1}(F) &= \tfrac{1}{2} f^{(0)+} (\mathcal{R}\mathbb{S})^{2n-1} \mathcal{R} f^{(0)} = \tfrac{1}{2} f^{(n)+} \mathbb{S}^{-1} f^{(n)} \\ &= \tfrac{1}{2}\alpha(F^{(n)}) = S_{-1}(F^{(n)}) \end{aligned} \tag{8.172}$$

corresponds to the polarizability with respect to the operator $F^{(n)}$. In cases where the operator $F^{(n)}$ has a simple form, we can find the corresponding sum rules very easily. In the last section we had examples of such cases.

8.7.5 Calculation of Transition Currents and Densities

A generalization of the usual sum rule (8.150) for the nuclear *density* (D.1) has been derived by Kao and Fallieros [KF 70, No 71, DF 73]:

$$S_\rho(\mathbf{r}) = \sum_\nu (E_\nu - E_0) \langle 0|\hat{\rho}(\mathbf{r})|\nu\rangle\langle\nu|F|0\rangle. \tag{8.173}$$

If we assume that F is a local, spin-independent operator of the form $F = \sum_i f(\mathbf{r}_i)$, it can be evaluated for velocity-independent forces:

$$S_\rho(\mathbf{r}) = \frac{1}{2}\langle 0|\big[\hat{\rho}(\mathbf{r}), \big[H, F\big]\big]|0\rangle = -\frac{\hbar^2}{2m}\nabla\big(\rho^{(0)}(\mathbf{r})\nabla f(\mathbf{r})\big), \tag{8.174}$$

where $\rho^{(0)}(\mathbf{r})$ is the density of the time reversal invariant ground state.

In the case where one state exhausts the total sum rule $S_\rho(\mathbf{r})$ (the "classical case"), $S_\rho(\mathbf{r})$ is proportional to the transition density (8.31) $\langle 0|\hat{\rho}(\mathbf{r})|\nu\rangle$. In such a case we have a connection between the quantum mechanical RPA calculation and the classical interpretation of a vibrating liquid drop, and we can study the actual form of the vibration which we have calculated in the RPA theory.

From Eqs. (8.173) and (8.174) we see that this form depends crucially on the transition operator F. It is determined by the requirement that *one* state exhausts a major part of the corresponding sum. As an example, we take

$$f(\mathbf{r}) = r^\lambda Y_{\lambda 0} \quad \text{for } \lambda > 0 \qquad \text{and} \qquad f(\mathbf{r}) = r^2 \quad \text{for } \lambda = 0.$$

In this case, we get

$$\rho^{(\nu)}(\mathbf{r}) = \langle 0|\hat{\rho}(\mathbf{r})|\nu\rangle \propto \begin{cases} \lambda r^{\lambda-1} Y_{\lambda 0} \dfrac{d}{dr}\rho^{(0)}, & \lambda \neq 0, \\ 3\rho^{(0)} + r\dfrac{d}{dr}\rho^{(0)}, & \lambda = 0. \end{cases} \tag{8.175}$$

We have seen that with these transition operators the giant resonances exhaust to a large extent the sum rules. We then get transition densities (8.175) which correspond exactly to hydrodynamic models, which will be discussed in Section 13.3. In the case of $\lambda > 0$, we get the *Bohr–Tassie model* [Bo 52, Ta 56a] of an incompressible irrotational fluid, and in the case $\lambda = 0$ we get the *Werntz–Überall model* [WU 66] of a compressible irrotational fluid.

Figure 8.14 shows transition densities for the case $\lambda = 0$ and $\lambda = 2$ in the nucleus ^{208}Pb. The simple expression (8.175) (dotted curves) are in very good agreement with the transition densities obtained from a continuum RPA calculation with the Skyrme I force (full lines). This shows that the giant resonances ($\lambda > 0$) to a large extent correspond to the surface vibrations of a liquid drop discussed in Chapter 1.

Suzuki and Rowe [SR 76, SR 77] derived a similar sum rule for the current density

$$\hat{\mathbf{j}}(\mathbf{r}) = \frac{\hbar}{2mi} \sum_{i=1}^{A} (\delta(\mathbf{r}-\mathbf{r}_i)\nabla_i + \nabla_i \delta(\mathbf{r}-\mathbf{r}_i)), \tag{8.176}$$

$$\begin{aligned} \mathbf{S}_\mathbf{j}(\mathbf{r}) &= \sum_\nu \langle 0|\hat{\mathbf{j}}(\mathbf{r})|\nu\rangle\langle\nu|F|0\rangle \\ &= \frac{1}{2mi}\rho^{(0)}(\mathbf{r})\nabla f(\mathbf{r}). \end{aligned} \tag{8.177}$$

In the "classical" case, where one state exhausts the entire sum, we can calculate the *transition current density*

$$\mathbf{j}^{(\nu)}(\mathbf{r}) = \langle 0|\hat{\mathbf{j}}(\mathbf{r})|\nu\rangle \propto \rho^{(0)}(\mathbf{r})\nabla f(\mathbf{r}). \tag{8.178}$$

This is the current which is connected to the collective motion in the same sense as the transition density (8.30) corresponds to the change in the density in the collective motion (see Sec. 13.3). We then can also define the

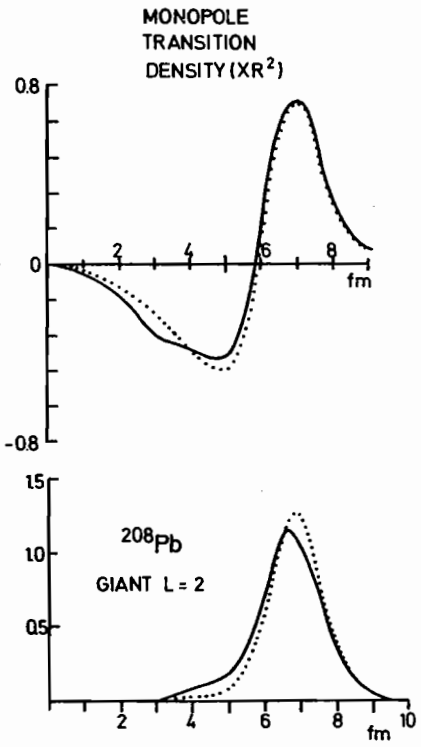


Figure 8.14. Isoscalar monopole and quadrupole transition densities in ^{208}Pb. (From [BT 75].)

velocity field

$$\mathbf{v}^{(\nu)}(\mathbf{r}) = \frac{\langle 0|\hat{\mathbf{j}}(\mathbf{r})|\nu\rangle}{\rho^{(0)}(\mathbf{r})} \propto \nabla f(\mathbf{r}). \tag{8.179}$$

It is irrotational—

$$\nabla \times \mathbf{v}^{(\nu)} = 0 \tag{8.180}$$

—for arbitrary, local one-body operators F. Furthermore, in the case of $f \sim r^{\lambda} Y_{\lambda 0}$, we get incompressibility (1.25) viz:

$$\nabla \mathbf{v}^{(\nu)} = 0.$$

We have to emphasize, however, that the simple classical picture of a local transition operator, which exhausts the entire sum rule, is only an approximation. Usually a part of the total strength is distributed also over some low-lying collective states (see Fig. 8.12).

In general, the excited state $|\nu\rangle$ can be written in the form*

$$|\nu\rangle = O_\nu^+ |0\rangle \qquad \text{with } O_\nu|0\rangle = 0,$$

* As long as we allow a general operator for O_ν^+, the following considerations are exact [St 79]. If we restrict ourselves to one-particle operators O_ν^+, we end up with the RPA approach.

and we obtain for the wave packet (8.27) in the limit of small amplitudes ($\epsilon \ll 1$):

$$|\Psi(t)\rangle = |0\rangle + \epsilon O_\nu^+ |0\rangle e^{-i\Omega_\nu t}$$
$$= e^{iA(t)}|0\rangle$$

with the Hermitian operator

$$A(t) = -i\epsilon\left(O_\nu^+ e^{-i\Omega_\nu t} - O_\nu e^{i\Omega_\nu t}\right).$$

We can also express $|\Psi(t)\rangle$ by the generalized coordinate $\mathcal{Q}_\nu$ and the momentum $\mathcal{P}_\nu$ as defined in Eq. (8.95), with $\epsilon' = \epsilon(2/M_\nu\Omega_\nu\hbar)^{1/2}$:

$$|\Psi(t)\rangle = \exp\left[-i\epsilon'(\Omega_\nu M_\nu \mathcal{Q}_\nu \sin\Omega_\nu t + \mathcal{P}_\nu \cos\Omega_\nu t)\right]|0\rangle.$$

If the ground state $|0\rangle$ and the operator O_ν^+ is time even,

$$TO_\nu^+ T^+ = O_\nu^+ \qquad \text{and} \qquad T|0\rangle = |0\rangle,$$

this representation corresponds to the separation of time-even ($\mathcal{Q}_\nu$) and time-odd parts ($\mathcal{P}_\nu$) in $A(t)$. They induce physically different effects in the nucleus: The transformation $\exp(-i\epsilon'\mathcal{P}_\nu \cos\Omega_\nu t)$ is time even. It causes "static"* deformations. On the contrary, the operation $\exp(-i\epsilon' M_\nu\Omega_\nu\mathcal{Q}_\nu \sin\Omega_\nu t)$ is time odd and induces currents. Both effects oscillate with the frequency Ω_ν.

The expectation value of any operator $\hat{T}$ in the state $|\Psi(t)\rangle$ is given by:

$$\langle\Psi(t)|\hat{T}|\Psi(t)\rangle = \langle 0|\hat{T}|0\rangle - i\epsilon'\langle 0|\left[\hat{T}, \mathcal{Q}_\nu\right]|0\rangle M_\nu\Omega_\nu \sin\Omega_\nu t$$
$$- i\epsilon'\langle 0|\left[\hat{T}, \mathcal{P}_\nu\right]|0\rangle \cos\Omega_\nu t.$$

Considering for the operator $\hat{T}$ either the density $\hat{\rho}(\mathbf{r})$ or the current $\hat{\mathbf{j}}(\mathbf{r})$, we obtain for the transition density and the transition current:

$$\rho^\nu(\mathbf{r}) = -\frac{i}{\hbar}\sqrt{\frac{\hbar}{2M_\nu\Omega_\nu}}\ \langle 0|\left[\hat{\rho}(\mathbf{r}), \mathcal{P}_\nu\right]|0\rangle,$$

$$\mathbf{j}^\nu(\mathbf{r}) = \sqrt{\frac{M_\nu\Omega_\nu}{2\hbar}}\ \langle 0|\left[\hat{\mathbf{j}}(\mathbf{r}), \mathcal{Q}_\nu\right]|0\rangle,$$

respectively. The state $|\nu\rangle$ is completely determined by the "coordinate" $\mathcal{Q}_\nu$. The corresponding "momentum" $\mathcal{P}_\nu$ can be calculated from Eq. (8.98). As an example, we obtain for a velocity independent single-particle operator

$$\mathcal{Q}_\nu = \sum_i f(\mathbf{r}_i),$$

and for a local two-body interaction the corresponding momentum

$$\mathcal{P}_\nu = \frac{1}{2}\frac{M_\nu}{m}\sum_i (\hat{\mathbf{p}}_i \nabla f(\mathbf{r}_i) + \text{h.c.}).$$

* We use here the word "static" in the sense of Section 12.3.1.

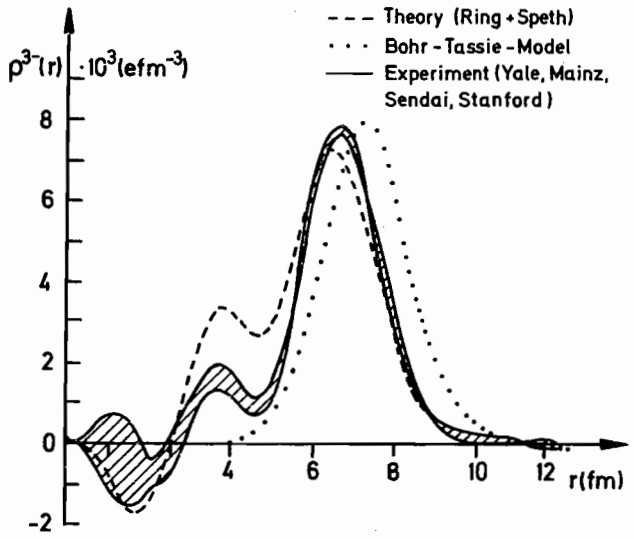


Figure 8.15. Theoretical and experimental radial transition density of the lowest 3^- state of ^{208}Pb. The full lines indicate the experimental errors [RFM 74], the dashed line [TW 73] is based on RPA calculations with the Migdal force, and the dotted line corresponds to the Bohr–Tassie model. (From [St 79].)

The constant M_ν is determined from the commutation relation (8.96). Again the velocity field is irrotational.

In general, we do not know $\mathfrak{D}_\nu$ a priori and obtain this operator only after the solution of the RPA equation. It then turns out that the low-lying collective states have large nonlocal components in $\mathfrak{D}$ [St 79]. They cannot be represented by the simple hydrodynamical models discussed above.

An example is the low-lying collective 3^- state in ^{208}Pb. Figure 8.15 shows experimental and theoretical transition densities $\rho^{3^-}(r)$. The experimental values deviate rather dramatically from the predictions of the Bohr–Tassie model. The fully microscopic calculation obtained by the solution of the RPA equation with a density dependent force is in much better agreement. Again for the high lying 3^- state the agreement with the Bohr–Tassie model is good [St 79].

8.8 Particle–Particle RPA

8.8.1 The Formalism

As in the Tamm–Dancoff case, where we distinguished between *ph* and *pp* TDA (Sec. 8.2.3), we have a *pp* RPA corresponding to the *ph* RPA. The *pp* TDA and the *pp* RPA give us states in the $A \pm 2$ systems. If they are collective, they are called *pairing vibrations* (see Sec. 8.3.5). Since many formal features of the *pp* RPA will be almost identical to those in the *ph* RPA, we will be quite brief in this section.

Corresponding to the *ph*-ansatz (8.65), in the *pp* case for the eigenstate τ of the $A+2$ system we have

$$|A+2,\tau\rangle=\Big(\sum_{m<n} X^{\tau}_{mn} a_m^+ a_n^+ - \sum_{i<j} Y^{\tau}_{ij} a_j^+ a_i^+\Big)|A,0\rangle$$
$$= R_\tau^+ |A,0\rangle. \tag{8.181}$$

The ground state (belonging to the A-system) is defined by

$$R_\tau |A,0\rangle = 0$$

and we get in the same approximation as in Eq. (8.69):

$$X^{\tau}_{mn} = \langle A,0|a_n a_m|A+2,\tau\rangle; \qquad Y^{\tau}_{ij} = \langle A,0|a_i a_j|A+2,\tau\rangle. \tag{8.182}$$

Proceeding as in the *ph* case (Sec. 8.4.1), and using the equation of motion for R_τ^+, we obtain* [FIS 64, Sch 64b, MLE 67, Ro 68a+b, RP 69]

$$\begin{aligned}(\hbar\Omega_\tau - \epsilon_m - \epsilon_n) X^{\tau}_{mn} &= \sum_{m'<n'} \bar{v}_{mnm'n'} X^{\tau}_{m'n'} - \sum_{i'<j'} \bar{v}_{mni'j'} Y^{\tau}_{i'j'},\\ (-\hbar\Omega_\tau + \epsilon_i + \epsilon_j) Y^{\tau}_{ij} &= -\sum_{m'<n'} \bar{v}_{ijm'n'} X^{\tau}_{m'n'} + \sum_{i'<j'} \bar{v}_{iji'j'} Y^{\tau}_{i'j'}.\end{aligned} \tag{8.183}$$

$\hbar\Omega_\tau = E_\tau^{A+2} - E_0^A$ is the excitation energy of the $A+2$ nucleus related to the ground state of the A-system.

We get identical equations for the $A-2$ particle system

$$\begin{aligned}(-\hbar\Omega_\lambda + \epsilon_i + \epsilon_j) X^{\lambda}_{ij} &= \sum_{i'<j'} \bar{v}_{iji'j'} X^{\lambda}_{i'j'} - \sum_{n'<m'} \bar{v}_{ijn'm'} Y^{\lambda}_{n'm'},\\ (\hbar\Omega_\lambda - \epsilon_n - \epsilon_m) Y^{\lambda}_{nm} &= -\sum_{i'<j'} \bar{v}_{nmi'j'} X^{\lambda}_{i'j'} + \sum_{n'<m'} \bar{v}_{nmn'm'} Y^{\lambda}_{n'm'},\end{aligned} \tag{8.184}$$

with $\hbar\Omega_\lambda = E_0^A - E_\lambda^{A-2}$,

$$X^{\lambda}_{ij} = \langle A-2,\lambda|a_j a_i|A,0\rangle, \qquad Y^{\lambda}_{nm} = \langle A-2,\lambda|a_m a_n|A,0\rangle. \tag{8.185}$$

We can therefore combine Eqs. (8.183) and (8.184) into one matrix equation similar to that in the *ph* case:

$$\begin{pmatrix} A & B \\ B^+ & C \end{pmatrix} \begin{bmatrix} R_p^{\tau,\lambda} \\ R_h^{\tau,\lambda} \end{bmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{bmatrix} R_p^{\tau,\lambda} \\ R_h^{\tau,\lambda} \end{bmatrix} \cdot \hbar\Omega_{\tau,\lambda}, \tag{8.186}$$

with

$$\begin{aligned} A_{mnm'n'} &= \delta_{mm'}\delta_{nn'}(\epsilon_m + \epsilon_n) + \bar{v}_{mnm'n'},\\ C_{iji'j'} &= -\delta_{ii'}\delta_{jj'}(\epsilon_i + \epsilon_j) + \bar{v}_{iji'j'},\\ B_{mnij} &= -\bar{v}_{mnij}, \end{aligned}$$

*The Eqs. (8.183) can also be derived from the time-dependent HF Eq. (8.116) generalized to include pair correlations. In application to nuclear physics this theory has been set up by H. Schmidt [Sch 64a+b].

and

$$\begin{aligned} (R_p^\tau)_{mn} &= X_{mn}^\tau, \qquad (R_p^\lambda)_{mn} = Y_{mn}^\lambda, \\ (R_h^\tau)_{ij} &= Y_{ij}^\tau, \qquad (R_h^\lambda)_{ij} = X_{ij}^\lambda. \end{aligned} \tag{8.187}$$

Note that, in general, the hermitian matrices A and C have different dimensions and that B is a rectangular matrix.

Equation (8.187) thus simultaneously yields the eigenstates of the $A \pm 2$ system, which are different from the *ph* case in which we had only solutions belonging to the same particle number. This, however, does not mean that the particle number is violated in *pp* RPA as it is in the BCS model (Chap. 6). The ansatz (8.181) conserves particle number. In practice, we recognize the solutions of Eq. (8.186), which belong to the *pp* system according to their wave functions: They have large *pp* components X_{mn} and small *hh* components Y_{ij}, and energies in the region of the *pp*-energies $\epsilon_m + \epsilon_n$. In cases where the *pp* and the *hh* components are of the same order of magnitude, we are, as in the *ph*-RPA, close to a phase transition, and the RPA approximation is not justified in any case.

Most of the other RPA features remain the same as in the *ph*-case. We list here, without proof, the normalization conditions:

$$\begin{aligned} \sum_{m<n} X_{mn}^{\tau*} X_{mn}^{\tau'} - \sum_{i<j} Y_{ij}^{\tau*} Y_{ij}^{\tau'} &= \delta_{\tau\tau'}, \\ \sum_{i<j} X_{ij}^{\lambda*} X_{ij}^{\lambda'} - \sum_{m<n} Y_{mn}^{\lambda*} Y_{mn}^{\lambda'} &= \delta_{\lambda\lambda'}, \end{aligned} \tag{8.188}$$

and the closure relation ($r<s$, $r'<s'$)

$$\sum_{\tau} X_{rs}^{\tau} X_{r's'}^{\tau*} - \sum_{\lambda} Y_{rs}^{\lambda*} Y_{r's'}^{\lambda} = \delta_{rr'}\delta_{ss'}. \tag{8.189}$$

The *pp*-RPA has been applied with success for closed shell $\pm$ 2-nucleons nuclei [VG 71, BV 71, KS 76]. It turns out, however, that in most cases a simple *pp* TDA calculation gives similar results. The $A \pm 2$ states are usually much less collective than the low-lying ph states. Therefore, the violation of the Pauli principle in the RPA approximation seems to be less justified than in the *ph* case. Therefore, the TDA approach is used in most of the $A \pm 2$ nuclei.

8.8.2 Ground State Correlations Induced by Pairing Vibrations

We have already seen that the ground state of the A-system is defined as the vacuum with respect to the boson operator O_ν (8.91), and in Eq. (8.101) we derived an explicit expression for it. By analogy, we now require that

$$R_\tau|\mathrm{RPA}\rangle = R_\lambda|\mathrm{RPA}\rangle = 0. \tag{8.190}$$

This yields an exponential factor of the form

$$|\mathrm{RPA}\rangle = N_0 \exp\Big(\sum_{m<n,\, i<j} Z_{mnij} A^+_{mn} A^+_{ij} \Big) |\mathrm{HF}\rangle, \tag{8.191}$$

with

$$A^+_{mn} = a^+_m a^+_n \,; \; A^+_{ij} = a_i a_j \qquad \text{and} \qquad Z^T = Y^* X^{*-1}.$$

If *pp* as well as *ph* correlations are important, the ground state has to be simultaneously the vacuum of the *ph* and *pp* bosons. We then gain an expression for the ground state having two exponential factors. These factors commute in the quasi-boson approximation.

Expanding Eq. (8.191), we see that the correlated ground state contains, besides the $|\mathrm{HF}\rangle$ state, contributions with two bosons, four bosons, and so on. The particle number is conserved, because the number of *pp* bosons is always equal to the number of the *hh* bosons.

To study the *pp* correlations in more detail, we restrict ourselves to a pure pairing force [see Eq. (4.140)] and a spherical closed shell nucleus. Since the pure pairing force is separable in the *pp* direction, we can again get the schematic model. In analogy to Eq. (8.135), we may derive from Eq. (8.184) the following dispersion relation.

$$\frac{2}{G} = \sum_n \frac{2j_n + 1}{2\epsilon_n - \hbar\Omega} + \sum_i \frac{2j_i + 1}{2\epsilon_i + \hbar\Omega}, \tag{8.192}$$

where n and i run over the particle and hole shells, respectively, and $2j+1$ is the number of levels in each of these shells. The single-particle energies are counted from the Fermi level, that is, $\epsilon_n > 0$, $\epsilon_i < 0$. The ground state has the form

$$|\mathrm{RPA}\rangle = N_0 \cdot \exp\Big(\sum_{ni} Z_{ni} A^+_n A^+_i \Big) |\mathrm{HF}\rangle = N_0 \sum_{\nu=0}^{\infty} \frac{1}{\nu!} \Big(\sum_{ni} Z_{ni} A^+_n A^+_i \Big)^{\nu} |\mathrm{HF}\rangle, \tag{8.193}$$

where $A^+ = \sum_{m>0} a^+_m a^+_{-m}$ is a 0^+ pair in the corresponding shell and the matrix Z is given by the equations

$$\sum_n Z_{nj} \frac{j_n + 1/2}{\hbar\Omega_\tau - 2\epsilon_n} = \frac{1}{\hbar\Omega_\tau + 2\epsilon_j} \qquad \text{for all } \tau \text{ values.}$$

For an increasing force constant, the number of virtual bosons becomes larger and we come to a phase transition as

$$G = G_c = 2 \Big(\sum_n \frac{2j_n + 1}{2\epsilon_n} + \sum_i \frac{2j_i + 1}{2\epsilon_i} \Big)^{-1}, \tag{8.194}$$

the excitation energy Ω goes to zero, and the RPA approach breaks down [Hö 61, BRS 68].

It is instructive to write down the BCS ground state in the form

$$|\mathrm{BCS}\rangle = N_0' \exp\Big(\sum_n \frac{v_n}{u_n} A^+_n + \sum_i \frac{u_i}{v_i} A^+_i \Big) |\mathrm{HF}\rangle. \tag{8.195}$$

It violates the particle number. If we project onto a good particle number, that is, if we neglect all components in which the number of *pp* bosons is different from the number of *hh* bosons, we get

$$|\text{PBCS}\rangle = N' \sum_{\nu=0}^{\infty} \frac{1}{\nu!}\frac{1}{\nu!}\left(\sum_{ni}\frac{v_n u_i}{u_n v_i} A_n^+ A_i^+\right)^\nu |\text{HF}\rangle. \tag{8.196}$$

On the other hand, we know that for small values of the force constant G, the gap equation (6.60)

$$\frac{2}{G} = \sum_n \frac{2j_n+1}{2\sqrt{\epsilon_n^2+\Delta^2}} + \sum_i \frac{2j_i+1}{2\sqrt{\epsilon_i^2+\Delta^2}} \tag{8.197}$$

has no solution ($\Delta = 0$). We find the same critical value G_c as in Eq. (8.194). For $G < G_c$ we use the correlated RPA ground state, and for $G > G_c$ we use the $|\text{BCS}\rangle$ state. Both approximations become poor in the vicinity of G_c. A way to avoid this problem would be a variation with respect to the occupation probabilities v^2 after projecting the particle number (see Chap. 11).

8.9 Quasi-particle RPA

Away from the closed shells, pair correlations become so important that the "feedback" of these correlations on the single-particle motion cannot be neglected any more. We have already seen in Chapters 6 and 7 that instead of $|\text{HF}\rangle$, the BCS or HFB ground state is appropriate in such cases. Since it is also a determinental wave function, we can construct TDA and RPA theory in complete analogy to the normal case* [Ba 60a]. It will thereby turn out that *ph* and *pp* RPA (TDA) may be encapsulated in one concise equation (see, for instance, [So 71]).

To derive these equations, we have to start with the quasi-particle representation (E.18)

$$H = H^0 + H^{11} + H^{31} + H^{40} + H^{22}. \tag{8.198}$$

Instead of Eq. (8.65), we now use the ansatz

$$Q_\nu^+ = \frac{1}{2}\sum_{kk'}(X^\nu_{kk'}\alpha_k^+\alpha_{k'}^+ - Y^\nu_{kk'}\alpha_{k'}\alpha_k) \tag{8.199}$$

and find, in analogy to the derivation of Section 8.4.1 the quasi-particle RPA equation whose matrix form is identical to Eq. (8.70). Only the matrix indices run through all pairs ($k < k'$) of the configuration space. In particular, this means *ph* indices (*mi*) and *pp*, and *hh* indices (*mn*), (*ij*). The

*The same equations have been derived by Bogoliubov within the Green's function formalism [Bo 59b].

matrices A and B are now given by $(k<k',\ l<l')$*

$$\begin{aligned} A_{kk'll'} &= \langle \mathrm{HFB} | \left[\alpha_{k'}\alpha_k, \left[H, \alpha_l^+ \alpha_{l'}^+ \right] \right] | \mathrm{HFB} \rangle \\ &= (E_k + E_{k'})\delta_{kl}\delta_{k'l'} + H^{22}_{kk'll'}, \\ B_{kk'll'} &= -\langle \mathrm{HFB} | \left[\alpha_{k'}\alpha_k, \left[H, \alpha_{l'}\alpha_l \right] \right] | \mathrm{HFB} \rangle = 4! \cdot H^{40}_{kk'll'}. \end{aligned} \tag{8.200}$$

Together they form the stability matrix (7.37) of the HFB theory, where we have used the representation of Eq. (E.18). In the canonical basis (in the case of a Nilsson + BCS calculation, this means the Nilsson basis; see Sec. 7.4) we find, for a time reversal invariant $|\mathrm{HFB}\rangle$ ground state with real matrix elements,

$$\begin{aligned} A_{kk'll'} = (E_k + E_{k'})\delta_{kl}\delta_{k'l'} &+ \bar{v}_{kk'll'}\tfrac{1}{2}(\xi^+_{kk'}\xi^+_{ll'} + \xi^-_{kk'}\xi^-_{ll'}) \\ &+ \bar{v}_{k\bar{l}'\bar{k}'l}\tfrac{1}{2}(\eta^+_{kk'}\eta^+_{ll'} + \eta^-_{kk'}\eta^-_{ll'}) \\ &- \bar{v}_{k\bar{l}\bar{k}'l'}\tfrac{1}{2}(\eta^+_{kk'}\eta^+_{ll'} - \eta^-_{kk'}\eta^-_{ll'}); \\ B_{kk'll'} = \quad & \bar{v}_{kk'\bar{l}\bar{l}'}\tfrac{1}{2}(\xi^+_{kk'}\xi^+_{ll'} - \xi^-_{kk'}\xi^-_{ll'}) \\ &+ \bar{v}_{kl\bar{k}'\bar{l}'}\tfrac{1}{2}(\eta^+_{kk'}\eta^+_{ll'} + \eta^-_{kk'}\eta^-_{ll'}) \\ &- \bar{v}_{kl'\bar{k}'\bar{l}}\tfrac{1}{2}(\eta^+_{kk'}\eta^+_{ll'} - \eta^-_{kk'}\eta^-_{ll'}); \end{aligned} \tag{8.201}$$

where

$$\xi^{\pm}_{kk'} = u_k u_{k'} \mp v_k v_{k'}; \qquad \eta^{\pm}_{kk'} = u_k v_{k'} \pm v_k u_{k'}. \tag{8.202}$$

In the HF-limit we get

$$\xi^{\pm}_{mi} = 0; \quad \xi^{\pm}_{mn} = 1; \quad \xi^{\pm}_{ij} = \mp 1; \quad \eta^{\pm}_{mi} = 1; \quad \eta^{\pm}_{mn} = \eta^{\pm}_{ij} = 0; \qquad \eta^{\pm}_{im} = \pm 1$$

which means that the set of *quasi-particle* RPA *equations* is decomposed, in this case, into the set of *ph* RPA equations (8.70), the set of *pp* RPA equations (8.186), and a set of *hh* RPA equations [the complex conjugate of (8.186)].

Furthermore, we see that the matrices A and B contain two types of matrix elements: *pp*-matrix elements $v_{kk'll'}$ multiplied by the factors ξ, which are close to one for *pp* or *hh* pairs; and the *ph* matrix elements $v_{k\bar{l}'\bar{k}'l}$ multiplied by factors η, which are close to one for *ph* pairs. In our derivation, both must be calculated with the same density-independent interaction. For density dependent interactions, however, we get the *ph* force [terms with η in Eq. (8.201)] as the derivative of the self-consistent field Γ with respect to the density ρ. The formalism of linear response theory can also be extended to the HFB case, working with the generalized densities $\mathcal{R}$ [Eq. (7.27)]. In this case, we get the *pp* force [terms with ξ in

* As we will see in Sec. 9.2.3.1, we can derive the quasi-particle RPA from a perturbative boson expansion, too. We have to be aware, however, that the matrix B obtained in Eq. (9.48) deviates from the matrix B in Eq. (8.200) by a factor of 3. This has its origin in the fact that higher order terms in the boson expansion are neglected [see also the discussion after Eq. (9.134)].

Eq. (8.201)] as a derivative of the pairing field Δ with respect to the pairing tensor κ [Mi 59, 60, ML 64, Mi 67, Bi 68, Ka 69, BEL 70].

The quasi-particle RPA equations have been widely used in connection with *separable forces*, such as the pairing-plus-quadrupole or the surface-delta interaction (see Secs. 4.4.6 and 4.4.7). As in the case of the schematic model (Sec. 8.5.2), the diagonalization problem is then reduced to a root search for relatively simple functions. This method provides a microscopic description of a large number of low-lying collective states, such as vibrations in spherical superconducting nuclei [Ar 63, ABV 63, AV 63, ASV 64]. It also has been successfully applied in the deformed rare earth and actinides nuclei for the description of β, γ, octupole, and pairing vibrations [SV 63, Be 63, So 65, FPM 67, MSU 67]. To each of these states belongs a rotational band, which can be calculated in the framework of the cranking model: We simply have to solve the RPA equations in a basis of rotating quasi-particles provided by a solution of the self-consistent cranking model as discussed in Section 7.7 [Ma 76b, MJ 78, EMR 80b]. Most of these calculations were concerned with the low-lying collective states. In recent years giant resonances in deformed regions in the framework of the quasi-particle RPA [KMY 75, ZS 75, MNS 76, ZS 76] were also investigated. Calculations with *realistic density-dependent forces* [ZSP 78] show again (as in spherical nuclei) that only a small fraction of the multipole sum rule is exhausted by the low-lying collective states. The giant resonances are much more collective and correspond to the classical surface oscillation (for instance, β or γ vibrations).